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47	Consistent*Complete*W	Algorithms
48	Microsoft Word - IE-EECMPE-COCHIN-17126-1397	Algorithms
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378	**44 COMMUNICATIONS OF THE ACM | AUGUST 2021 | VOL. 64 | NO. 8practice**	**explainable-ai**
379	**DECODING TRUST : A Comprehensive Assessment of**	**explainable-ai**
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380	Fairness in Machine Learning - 2019-09-19	fair-ml-book

381	Fairness in Machine Learning - 2019-09-19	fair-ml-book
382	Fairness in Machine Learning - 2019-09-22	fair-ml-book
fewshotlearning		
383	untitled	fewshotlearning
384	ArcFace: Additive Angular Margin Loss for Deep Face Recognition	fewshotlearning
385	Generalizing from a Few Examples: A Survey on Few-Shot Learning	fewshotlearning
history		
386	**Microsoft Word - TuringTest.doc**	**history**
387		history
language-understanding		
388	Corpus-Based Approaches to Semantic Interpretation in NLP	language-understanding
389	Dialogue Systems as Proof Editors	language-understanding
390	Journal of Artificial Intelligence Research 37 (2010) 141-188 Submitted 10/09; published 02/10	language-understanding
391	The Handbook of Computational Linguistics and Natural Language Processing	language-understanding
392	Recognition of Emotion from Speech: A Review	language-understanding
393	Unsupervised Relation Discovery with Sense Disambiguation	language-understanding
394	On our best behaviour	language-understanding
395	Recursive Deep Models for Semantic Compositionality Over a Sentiment Treebank	language-understanding
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397	My title	language-understanding
398	Learning Distributed Word Representations for Natural Logic Reasoning	language-understanding
399	**Beyond Accuracy: Behavioral Testing of NLP Models with C heckList**	**language-understanding**
400	Modeling Intra and Inter-modality Incongruity for Multi-Modal Sarcasm Detection	language-understanding
401	arXiv:2102.02503v1 [cs.CL] 4 Feb 2021Understanding the Capabilities, Limitations, and	language-understanding
machinelearning		
402	"! # $&%('*)+-,/.1012 %435+6' 78+9%($:,*);,=< > ?@?*AB)+6' .&C D CFEHG@I +-,/.J	machinelearning
403	1 Introduction to Semi-Supervised Learning	machinelearning
404	rosenblatt 1957	machinelearning
405	Data Clustering: A Review	machinelearning
406	19969_fraud_final.qxd	machinelearning
407	504 504..507	machinelearning
408	Com bining Lab eled and Unlab eled Data with Co/-T raining	machinelearning
409	8.LEARNING INTERNAL REPRESENT AnONS319	machinelearning
410	CONTENTS	machinelearning
411	Data Min Knowl Disc (2007) 15:55–86	machinelearning
412	doi:10.1016/j.ejor.2006.06.072	machinelearning
413	EXPLORATIONS IN	machinelearning

414	j1993adaptiveneural	machinelearning
415	c1960adaptiveswitchingcircuits deltarule	machinelearning
416	ftml.dvi	machinelearning
417	Semi-Supervised Learning	machinelearning
418	untitled	machinelearning
419	**201Why Does Unsupervised Pre-training Help Deep Learning?**	**machinelearning**
420	AnIntroductiontoMachineLearning	machinelearning
421	78 communications of the acm | october 2012 | vol. 55 | no. 10review articles	machinelearning
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423	**nc.dvi**	**machinelearning**
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425	An Interactive Node-Link Visualization	machinelearning
426	Deep Convolutional Networks on Graph-Structured	machinelearning
427	Efficient and Robust Automated Machine Learning	machinelearning
428	Hidden Technical Debt in Machine Learning Systems	machinelearning
429	Hidden Technical Debt in Machine Learning Systems	machinelearning
430	Transactions on Machine Learning	machinelearning
431	Google’s Neural Machine Translation System: Bridging the Gap	machinelearning

432	June 25th 2016SPECIAL REPORT	machinelearning
433	Machine learning and computer algebra	machinelearning
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435	**Attention Is All You Need**	**machinelearning**
436	CSUR5002-21	machinelearning
437	**Megapixel Size Image Creation using Generative Adversarial**	**machinelearning**
438	ATM: A distributed, collaborative, scalable system for automated machine learning	machinelearning
439	Model-based Pricing for Machine Learning in a Data Marketplace	machinelearning
440	Data Platform for Machine Learning	machinelearning
441	Towards Federated Learning at Scale: System Design	machinelearning
442	C h a p t e r 1 0	machinelearning
443	NeurIPS_2020[3].pdf	machinelearning
444	Underspecification in Machine Learning	machinelearning
445	Underspecification in Machine Learning	machinelearning
446	What is being transferred in transfer learning?	machinelearning
447	**Informer: Beyond Efficient Transformer for Long Sequence**	**machinelearning**
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449	Safe: Enhancing Mathematical Reasoning in Large Language Models via Retrospective Step-aware Formal Verification	machinelearning

machinetranslation		
450	mtcl90.pdf	machinetranslation
451	An Architecture For More Realistic Conversational	machinetranslation
452	**Microsoft Word - horvitze.doc**	**machinetranslation**
453	The Mathematics of Statistical Machine Translation: Parameter Estimation	machinetranslation
454	ACMB046B-08.tex	machinetranslation
455	Translation	machinetranslation
456	A Recursive Recurrent Neural Network for Statistical Machine Translation	machinetranslation
457	arXiv:1409.3215v3 [cs.CL] 14 Dec 2014	machinetranslation
458	A Neural Conversational Model	machinetranslation
459	Google's Neural Machine Translation System: Bridging the Gap	machinetranslation
460	Google's Neural Machine Translation System: Bridging the Gap	machinetranslation
461	Published as a conference paper at ICLR 2016	machinetranslation
462	Toward Multilingual Neural Machine Translation	machinetranslation
463	Neural Machine Translation and Sequence-to-sequence Models:	machinetranslation
464	Neural Machine Translation and Sequence-to-sequence Models:	machinetranslation
465	Sequence to Sequence Learning	machinetranslation
ml-and-optimization		
466	MLOPT-intro06a.dvi	ml-and-optimization

467	Constraints manuscript No.	ml-and-optimization
468	Learning to learn by gradient descent	ml-and-optimization
469	SIAM R EVIEW c2017 Society for Industrial and Applied Mathematics ○	ml-and-optimization
470	Accelerating Primal Solution Findings for Mixed Integer Programs	ml-and-optimization
471	arXiv:1906.01629v2 [cs.LG] 7 Jun 2019Exact Combinatorial Optimization	ml-and-optimization
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472	Efficient Deployment of Predictive Analytics through Open	operationalize
473	The Missing Piece in Complex Analytics: Low Latency,	operationalize
474	56 COMMUNICATIONS OF THE ACM | JULY 2018 | VOL. 61 | NO. 7contributed articles	operationalize
475	An artificial intelligence life cycle: From conception to production	operationalize
patterns		
476	Fingerprint Matching Algorithm Based on Tree	patterns
477	Microsoft Word - 05_11985 - IJECE__pp 279-289	patterns
478	Automated Inference on Criminality using Face Images	patterns
reinforcementlearning		
479	rl-survey.dvi	reinforcementlearning
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480	**Published as a conference paper at ICLR 2017**	**sharing-secrets**
481	5/23/2018 How will the GDPR impact machine learning? - O'Reilly Media	sharing-secrets
482	5/23/2018 Is there a 'right to explanation' for machine learning in the GDPR?	sharing-secrets
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483	Measure It? Manage It? Ignore It? Software Practitioners and Technical Debt	softwarenegineering
484	FullStack Bench: Evaluating LLMs as Full Stack Coders	softwarenegineering
485	debug-gym : A Text-Based Environment for	softwarenegineering
486	From Code Foundation Models to Agents and Applications: A Comprehensive Survey and Practical Guide to Code Intelligence	softwarenegineering
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487	TURING AWARD LECTURE	softwaresecurity
488	Andrea Bittau, Adam Belay, Ali Mashtizadeh, David Mazi `eres, Dan Boneh	softwaresecurity
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489	**Matrix Computations and Optimization in Apache Spark**	**spark**
490	Robust Large-Scale Machine Learning in the Cloud	spark
syntheticdata		
491	Synner: Generating Realistic Synthetic Data	syntheticdata
492	TinyTroupe: An LLM-powered Multiagent Persona Simulation Toolkit	syntheticdata

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Legend: ✓ = Landmark Paper (highlighted in gold) ☐ = Standard Paper

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Short Summary
1Last month, I got a phone call. Okay maybe that's not exactly how it happened, and maybe those weren't his exact words. But after learning about the new company Elon Musk was starting, I've come to realize that that's exactly what he's trying to do. When I wrote about Tesla and SpaceX , I learned that you can only fully wrap your head around certain
Hybrid learning methods use theoretical knowledge of a doma in and a set of classified examples to develop a method for accurately classifying exa mples not seen during training. The challenge of hybrid learning systems is to use the information provided by one source of information to offset information missing f rom the other source. By so doing, a
In International Journal Human-Computer Studies 43, p.907-928. Substantial revision of paper presented at the International Workshop on Formal Ontology, March, 1993, Padova, Italy. Available as Technical Repor t KSL 93-04, Knowledge Systems Laboratory, Stanford University. Toward Principles for the Design of Ontologies Used for
25th Anniversary Issue 68 AI MAGAZINE■I claim that achieving real human-level artificial intelligence would necessarily imply that most ofthe tasks that humans perform for pay could be au-tomated. Rather than work toward this goal of au-tomation by building special-purpose systems, I ar-gue for the development of general-purpose,educable systems that
In this paper, we show that in the best-known version of the bl ocks world (and several related versions), planning is difficult, in the sens e that finding an optimal plan is NP-hard. However, the NP-hardness is not due to deleted-c ondition interactions, but instead due to a situation which we call a deadlock. For probl ems that do not contain deadlocks, there
0.0 THE QUEST FOR ARTIFICIAL INTELLIGENCE A HISTORY OF IDEAS AND ACHIEVEMENTS Web Version Print version published by Cambridge University Press http://www.cambridge.org/us/0521122937 Nils J. Nilsson Stanford University Copyright c2010 Nils J. Nilsson http://ai.stanford.edu/ nilsson/ All rights reserved. Please do not reproduce
Criticisms of the Turing Test and Why You Should Ignore (Most of) Them Katrina LaCurts katrina@csail.mit.edu MIT CSAIL, 6.893 1 6.893 Katrina LaCurts 1 Introduction In this essay, I describe a variety of criticisms against using The Turing Test (from here on out, “TT”) as a test for machine intelligence. These criticisms range from
UNCLASSIFIED Ontologies and Information Systems: A Literature Survey Van Nguyen Command, Control, Communications and Intelligence Division Defence Science and Technology Organisation DSTO–TN–1002 ABSTRACT Anontology captures in a computer-processable language the important con- cepts in a particular domain and the
arXiv:1409.3215v3 [cs.CL] 14 Dec 2014Sequenceto SequenceLearning withNeuralNetworks IlyaSutskever Google ilyasu@google.comOriolVinyals Google vinyals@google.comQuocV.Le Google qvl@google.com Abstract Deep Neural Networks (DNNs) are powerful models that have ac hieved excel- lentperformanceondifficultlearningtasks.
Google’s Neural Machine Translation System: Bridging the Gap between Human and Machine Translation Yonghui Wu, Mike Schuster, Zhifeng Chen, Quoc V. Le, Mohammad Norouzi yonghui,schuster,zhifengc,qvl,mnorouzi@google.com Wolfgang Macherey, Maxim Krikun, Yuan Cao, Qin Gao, Klaus Macherey, Jeff Klingner, Apurva Shah, Melvin Johnson,
10 COMMUNICATIONS OF THE ACM | JANUARY 2018 | VOL. 61 | NO. 1letters to the editor chine be given the intelligence to make such a determination on its own? Health-care institutions increasing - ly must be able to identify authentically human patients’ EHRs. In light of this trend, hackers might want to generate synthetic patient identities to
60 COMMUNICATIONS OF THE ACM | SEPTEMBER 2017 | VOL. 60 | NO. 9THE FIELD OF artificial intelligence has made great strides recently, as in AlphaGo’s victories in the game of Go over world champion South Korean Lee Sedol in March 2016 and top-ranked Chinese Go player Ke Jie in May 2017, leading to great
82 COMMUNICATIONS OF THE ACM | JUNE 2017 | VOL. 60 | NO. 6Technical Perspective What Led Computer Vision to Deep Learning? By Jitendra Malik vincing the computer vision commu- nity would require results on the real- world datasets that we used. Geoff did take this advice to heart and I like to think that conversation
Massive Exploration of Neural Machine Translation Architectures Denny Britz∗†, Anna Goldie∗, Minh-Thang Luong, Quoc Le {dennybritz,agoldie,thangluong,qvl }@google.com Google Brain Abstract Neural Machine Translation (NMT) has shown remarkable progress over the past few years with production systems now being deployed to end-users.
70 COMMUNICATIONS OF THE ACM | NOVEMBER 2018 | VOL. 61 | NO. 11china region CHINA’S E-COMMERCE BOOM has generated a huge logistics demand, both in terms of express package delivery and on-demand food delivery.a Chinese express firms delivered an estimated 40 billion parcels in 2017; up 28% from the previous year.2
FEBRUARY 2018 | VOL. 61 | NO. 2 | COMMUNICATIONS OF THE ACM 7cerf’s up Every few years, I have to pass a test from the Department of Motor Vehicles to drive my car in Virginia (and the rest of the U.S.). Shouldn’t a self-driving car be required to do the same thing? Actually, the Waymo self-driving car passes a more comprehensive set of tests

The Secret Sharer: Measuring Unintended Neural Network Memorization & Extracting Secrets Nicholas Carlini University of California, BerkeleyChang Liu University of California, Berkeley Jernej Kos National University of Singapore´Ulfar Erlingsson Google BrainDawn Song University of California, Berkeley Abstract Machine learning models based on neural networks and deep learning are being rapidly ado

54 COMMUNICATIONS OF THE ACM | MARCH 2019 | VOL. 62 | NO. 3contributed articles ILLUSTRATION BY VAULT49THE DRAMATIC SUCCESS I n machine learning has led to an explosion of artificial intelligence (AI) applications and increasing expectations for autonomous systems that exhibit human-level intelligence. These expectations have, however, met with fundamental obstacles that cut across many applicati

AI models are increasingly applied in high-stakes domains like health and conservation. Data quality carries an elevated signifi- cance in high-stakes AI due to its heightened downstream impact, impacting predictions like cancer detection, wildlife poaching, and loan allocations. Paradoxically, data is the most under-valued and de-glamorised aspect of AI. In this paper, we report on data practices

2/3/22, 9:31 PM The Bitter Lesson www .incompleteideas.net/IncIdeas/BitterLesson.html 1/2The Bitter Lesson Rich Sutton March 13, 2019 The biggest lesson that can be r ead fr om 70 years of AI r esear ch is that general methods that leverage computation ar e ultimately the most ef fective, and by a lar ge mar gin. The ultimate r eason for this is Moor ele law, or rather its generalization of conti

Artificial intelligence (AI) researchers have been developing and refining large language models (LLMs) that exhibit remarkable capabilities across a variety of domains and tasks, challenging our understanding of learning and cognition. The latest model developed by OpenAI, GPT-4 [Ope23], was trained using an

As of December 2024, the ARC-AGI benchmark is five years old and remains unbeaten. We believe it is currently the most important unsolved AI benchmark in the world because it seeks to measure generalization on novel tasks – the essence of intelligence – as opposed to skill at tasks that can be prepared for in advance. This year, we launched ARC Prize, a global competition to inspire new ideas and

Leopold Aschenbrenner S I T U AT I O N A L AWA R E N E S S The Decade Ahead JUNE 2024 2 Dedicated to Ilya Sutskever. While I used to work at OpenAI, all of this is based on publicly- available information, my own ideas, general field-knowledge, or SF-gossip. Thank you to Collin Burns, Avital Balwit, Carl Shulman, Jan Leike, Ilya Sutskever, Holden Karnofsky, Sholto Douglas, James Bradbury, Dwarkesh

AI models excel at creating content, but typically render it with static, predefined interfaces. Specifically, the output of LLMs is often a markdown “wall of text”. Generative UI is a long standing promise, where the model generates not just the content, but the interface itself. Until now, Generative UI was not possible in a robust fashion. We demonstrate that when properly prompted and equipped w

Generative UI is transforming interface design by facilitating AI- driven collaborative workflows between designers and computa- tional systems. This study establishes a working definition of Gen- erative UI through a multi-method qualitative approach, integrating insights from a systematic literature review of 127 publications, ex- pert interviews with 18 participants, and analyses of 19 case stu

Generative UI is transforming interface design by facilitating AI- driven collaborative work!ows between designers and computa- tional systems. This study establishes a working de"nition of Gen- erative UI through a multi-method qualitative approach, integrating insights from a systematic literature review of 127 publications, ex- pert interviews with 18 participants, and analyses of 19 case studi

arXiv:2506.18096v1 [cs.AI] 22 Jun 2025DEEPRESEARCH AGENTS : A S YSTEMATIC EXAMINATION ANDROADMAP Yuxuan Huang†1, Yihang Chen†2, Haozheng Zhang†2, Kang Li3, Meng Fang1, Linyi Yang4, Xiaoguang Li2, Lifeng Shang2, Songcen Xu2, Jianye Hao2, Kun Shao‡2, and Jun Wang‡4 1University of Liverpool2Huawei Noah’s Ark Lab3University of Oxford4University College London ABSTRACT The rapid progress of Large Langu

A Perspective on Decentralizing AI Abhishek Singh, Charles Lu, Gauri Gupta, Nikhil Behari, Ayush Chopra, Jonas Blanc, Tzofi Klinghoffer, Kushagra Tiwary, and Ramesh Raskar MIT Media Lab 1.Introduction Current progress in AI is dominated by a few large organizations with centralized data and compute resources. However, as AI expands into real-world industries, such as healthcare, finance, supply chain

Adaptation of Agentic AI Pengcheng Jiang1∗, Jiacheng Lin1∗, Zhiyi Shi1,4∗, Zifeng Wang13, Luxi He3, Yichen Wu4, Ming Zhong1, Peiyang Song5,6, Qizheng Zhang2, Heng Wang1, Xueqiang Xu1, Hanwen Xu7, Pengrui Han1, Dylan Zhang1, Jiashuo Sun1, Chaoqi Yang1, Kun Qian14, Tian Wang14, Changran Hu5, Manling Li10, Quanzheng Li4,12, Hao Peng1, Sheng Wang7, Jingbo Shang8, Chao Zhang9, Jiaxuan You1, Liyuan Liu1

The endowment of AI with reasoning capabilities and some degree of agency is widely viewed as a path toward more capable and generalizable sys- tems. Our position is that the current development of agentic AI requires a more holistic, systems- theoretic perspective in order to fully understand their capabilities and mitigate any emergent risks. The primary motivation for our position is that AI de

—This review critically distinguishes between AI Agents and Agentic AI, offering a structured conceptual tax- onomy, application mapping, and challenge analysis to clarify their divergent design philosophies and capabilities. We begin by outlining the search strategy and foundational definitions, characterizing AI Agents as modular systems driven by LLMs and LIMs for narrow, task-specific automati

AI Agents: Evolution, Architecture, and Real-World Applications Naveen Krishnan March 2025 1 Introduction Artificial Intelligence (AI) has evolved dramatically over the past decade, tran- sitioning from specialized systems designed for narrow tasks to increasingly so- phisticated architectures capable of autonomous operation across diverse do- mains. Among these advancements, AI agents represent a

As AI-driven agents become increasingly integrated into the digital ecosystem, they reshape how online advertising is perceived and processed. Particularly in the travel and hotel booking sector, these autonomous systems influence the effectiveness of traditional advertising formats. While visual cues and emotional appeals sway human users, AI agents prioritize structured data, such as price, avail

Designing LLM-based Multi-Agent Systems for Software Engineering Tasks: Quality Attributes, Design Patterns and Rationale YANGXIAO CAI,School of Computer Science, Wuhan University, China RUIYIN LI,School of Computer Science, Wuhan University, China PENG LIANG,School of Computer Science, Wuhan University, China MOJTABA

From Code Foundation Models to Agents and Applications: A Comprehensive Survey and Practical Guide to Code Intelligence BUAA-SKLCCSE, Alibaba, ByteDance, M-A-P , BJTU, OPPO, HKUST (GZ), BUPT, TeleAI, Shanghai AI Lab, Manchester, StepFun, UoS, SCU, CASIA, NJU, Kuaishou, HIT, Huawei Cloud, Tencent, Monash/CSIRO, NTU,

AI agents are already operating in production across many industries, yet there is limited pub- lic understanding of the technical strategies that make deployments successful. We present the first large-scale systematic study of AI agents in production, surveying 306 practitioners and con- ducting 20 in-depth case studies via interviews across 26 domains. We

towards a science of scaling agent systems yubin kimQLSL ken guQL chanwoo parkSL chunjong parkRL samuel schmidgallRL aN ali heydariQL yao yanQL zhihan zhangQL yuchen zhuangRL mark malhotraQL paul pu liangSL hae won parkSL yuzhe yangQL xuhai xuQL yilun duQL shwetak patelQL tim altho!QL daniel mcdu!Qand xin liuQ Qgoo©le

. . Latest updates: h□ps://dl.acm.org/doi/10.1145/3776528 . . SURVEY A Survey of AI Reliance SVEN ECKHARDT, University of Zurich, Zurich, ZH, Switzerland . NIKLAS KÜHL, University of Bayreuth, Bayreuth, Bayern, Germany . MATEUSZ DOLATA, University of Zurich, Zurich, ZH, Switzerland . GERHARD SCHWABE, University of Zurich, Zurich,

of Group-Evolving Agents (GEA) vs. tree-structured self-evolution for open-endedness.GEA treats agroup of agents, rather than an individual agent, as the fundamental unit of evolution. At each iteration, a parent group jointly gives rise to an offspring group through explicit intra-group Experience sharing and reuse.

An Improved Data Stream Summary: The Count-Min Sketch and its Applications Graham Cormode⋆and S. Muthukrishnan⋆⋆ Abstract. We introduce a new sublinear space data structure—the Count-Min Sketch— for summarizing data streams. Our sketch allows fundamental queries in data stream summarization such as point, range,

Appr oximatel yDetecting Duplicates forStreaming Data using StableBloom Filter s FanDeng University ofAlberta fandeng@cs .ualber ta.caDavoodRafiei University ofAlberta drafiei@cs .ualber ta.ca ABSTRA CT Traditional duplicate elimination techniques arenotappli- cable tomanydatastream applications. Ingeneral, pre- cisely eliminating duplicates

BloomFiltersprovidespace-efficientstorageofsets at the cost of a probability of false positives on mem- bershipqueries. Thesizeofthefiltermustbedefined a prioribasedonthenumberofelementstostoreandthe desired false positive probability, being impossible to store extra elements without increasing the false posi- tive probability. This leads typically to a

Knowl Inf Syst (2008) 14:1–37 DOI 10.1007/s10115-007-0114-2 SURVEY PAPER Top 10 algorithms in data mining Xindong Wu ·Vipin Kumar ·J. Ross Quinlan ·Joydeep Ghosh ·Qiang Yang · Hiroshi Motoda ·Geoffrey J. McLachlan ·Angus Ng ·Bing Liu ·Philip S. Yu · Zhi-Hua Zhou ·Michael Steinbach ·David J. Hand ·Dan Steinberg Received: 9 July

describes and analyses a near-optimal probabilistic algorithm, H YPER LOGLOG, dedicated to estimating the number of distinct elements (the cardinality) of very large data ensembles. Using an auxiliary memory ofmunits (typically, “short bytes”), H YPER LOGLOGperforms a single pass over the data and produces an estimate of the cardinality such that

Foundations and TrendsRin DatabasesVol. 4, Nos. 1–3 (2011) 1–294 c2012 G. Cormode, M. Garofalakis, P. J. Haas and C. JermaineDOI: 10.1561/1900000004 Synopses for Massive Data: Samples, Histograms, Wavelets, Sketches By Graham Cormode, Minos Garofalakis, Peter J. Haas and Chris Jermaine Contents 1 Introduction 2 1.1 The Need for

A Machine Learning Approach to Databases Indexes Alex Beutel, Tim Kraska∗, Ed H. Chi, Jeffrey Dean, Neoklis Polyzotis Google, Inc. Mountain View, CA {alexbeutel,kraska,edchi,jeff,npolyzotis}@google.com Abstract Databases rely on indexing data structures to efficiently perform many of their core operations. In order to look up all records in a

Consistent*Complete*W ellDocumented*EasytoReuse* *Evaluated *CAV*Artifact* AECReluplex: An Efficient SMT Solver for Verifying Deep Neural Networks⋆ Guy Katz, Clark Barrett, David Dill, Kyle Julian and Mykel Kochenderfer Stanford University, USA {guyk, clarkbarrett, dill, kjulian3, mykel }@stanford.edu Abstract. Deep neural networks have

of Hashing Techniques Using Neural Network

The Case for Learned Index Structures Tim Kraska∗ MIT Cambridge, MA kraska@mit.eduAlex Beutel Google, Inc. Mountain View, CA alexbeutel@google.comEd H. Chi Google, Inc. Mountain View, CA edchi@google.com Jeffrey Dean Google, Inc. Mountain View, CA jeff@google.comNeoklis Polyzotis Google, Inc. Mountain View, CA

Building a scalable and real-time recommendation system is vital for many businesses driven by time-sensitive customer feedback, such as short-videos ranking or online ads. Despite the ubiquitous adoption of production-scale deep learning frameworks like Ten- sorFlow or PyTorch, these general-purpose frameworks fall short of business

In recent years, an active field of research has developed around automated machine learning (AutoML). Unfortunately, comparing different AutoML systems is hard and often done incorrectly. We introduce an open, ongoing, and extensible benchmark framework which follows best practices and avoids common mistakes. The framework is open-source, uses

With rapidly advancing technologies like robotic process automation (RPA), artificial intelligence (AI), a nd machine learning (ML), businesses are automating processes at unprecedented rates. Hyperautomation represents the next evolution of business process automation, combining these tools to create end -to-end automated systems. This paper

See discussions, st ats, and author pr ofiles f or this public ation at : https://www .researchgate.ne t/public ation/26501575 A Su rvey of Homomorphic Encryption for Nonspecialists Article in EUR ASIP Journal on Inf ormation Security · Januar y 2007 DOI: 10.1155/2007/13801 · Sour ce: DO AJ CITATIONS 311READS 448 2 author s, including:

79A Survey on Homomorphic Encryption Schemes: Theory and Implementation ABBASACAR,HIDAYETAKSU,and A.SELCUK ULUAGAC, FloridaInternationalUniversity MAUROCONTI, Universityof Padua

The capacity of a neural network to absorb information is limited by its number of parameters. Conditional computation, where parts of the network are active on a per-example basis, has been proposed in theory as a way of dramatically increas- ing model capacity without a proportional increase in computation. In practice, however, there are significant

We study empirical scaling laws for language model performance on the cross-entropy loss. The loss scales as a power-law with model size, dataset size, and the amount of compute used for training, with some trends spanning more than seven orders of magnitude. Other architectural details such as network width or depth have minimal effects within a

arXiv:2501.09223v2 [cs.CL] 15 Jun 2025Foundations of Large Language Models Tong Xiao and Jingbo Zhu June 17, 2025 NLP Lab, Northeastern University & NiuTrans Research This book is a selection of chapters from an introductory NLP resource available at https://github.com/NiuTrans/NLPBook Copyright © 2021-2025 Tong Xiao and Jingbo Zhu

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks Patrick Lewis†‡, Ethan Perez⋆, Aleksandra Piktus†, Fabio Petroni†, Vladimir Karpukhin†, Naman Goyal†, Heinrich Küttler†, Mike Lewis†, Wen-tau Yih†, Tim Rocktäschel†‡, Sebastian Riedel†‡, Douwe Kiela† †Facebook AI Research;‡University College London;⋆New York

—Prompt engineering is an increasingly important skill set needed to converse effectively with large languag e models (LLMs), such as ChatGPT. Prompts are instructions given to a n LLM to enforce rules, automate processes, and ensure specifi c qualities (and quantities) of generated output. Prompts ar e also a form of programming that can customize the

The use of NLP in the realm of financial technology is broad and complex, with applications ranging from sentiment analysis and named entity recognition to question answering. Large Language Models (LLMs) have been shown to be effective on a variety of tasks; however, no LLM specialized for the financial domain has been

Chain-of-Thought Prompting Elicits Reasoning in Large Language Models Jason Wei Xuezhi Wang Dale Schuurmans Maarten Bosma Brian Ichter Fei Xia Ed H. Chi Quoc V . Le Denny Zhou Google Research, Brain Team {jasonwei,dennyzhou}@google.com Abstract We explore how generating a chain of thought —a series of intermediate

Teaching LLMs to Plan: Logical Chain-of-Thought Instruction Tuning for Symbolic Planning Pulkit Verma MIT CSAIL Cambridge, USA pulkitv@mit.eduNgoc La MIT CSAIL Cambridge, USA ntmla@mit.eduAnthony Favier MIT CSAIL Cambridge, USA antfav24@mit.edu Swaroop Mishra∗ Microsoft AI Mountain View, USA

GenerativeAgentSimulationsof1,000People Authors: Joon Sung Park1*,Carolyn Q.Zou1,2,Aaron Shaw2,Benjamin Mako Hill3,Carrie Cai4, Meredith Ringel Morris5,Robb Willer6,Percy Liang1,Michael S.Bernstein1 Affiliations: 1Computer Science Department, Stanford University; Stanford, CA, 94305, USA. 2Department ofCommunication Studies,

The rapid development of large language models has revolutionized code intelligence in software development. However, the predominance of closed-source models has restricted extensive research and development. To address this, we introduce the DeepSeek-Coder series, a range of open-source code models with sizes from 1.3B to 33B,

GSM-Symbolic: Understanding the Limitations of Mathematical Reasoning in Large Language Models Iman Mirzadeh†Keivan Alizadeh Hooman Shahrokhi∗ Oncel Tuzel Samy Bengio Mehrdad Farajtabar† Apple Abstract Recent advancements in Large Language Models (LLMs) have sparked interest in their formal reasoning

Large Language Models (LLMs) have shown useful applications in a variety of tasks, including data wrangling. In this paper, we investigate the use of an off-the-shelf LLM for schema matching. Our objective is to identify semantic correspondences between elements of two relational schemas using only names and descriptions. Using a newly

Recent research, such as BitNet [WMD+23], is paving the way for a new era of 1- bit Large Language Models (LLMs). In this work, we introduce a 1-bit LLM variant, namely BitNet b1.58 , in which every single parameter (or weight) of the LLM is ternary {-1, 0, 1}. It matches the full-precision (i.e., FP16 or BF16) Transformer LLM with the same model size

The Llama 3 Herd of Models Llama Team, AI @ Meta1 1A detailed contributor list can be found in the appendix of this paper. Modern artificial intelligence (AI) systems are powered by foundation models. This paper presents a new set of foundation models, called Llama 3. It is a herd of language models that natively support multilinguality, coding,

A Survey of AI Agent Protocols Yingxuan Yang, Huacan Chai, Yuanyi Song, Siyuan Qi, Muning Wen, Ning Li, Junwei Liao, Haoyi Hu, Jianghao Lin→, Gaowei Chang†, Weiwen Liu, Ying Wen, Yong Yu, Weinan Zhang Shanghai Jiao Tong University,†ANP Community {zoeyyx, chiangel, wnzhang}@sjtu.edu.cn Abstract The rapid development of large

ADVANCES AND CHALLENGES IN FOUNDATION AGENTS FROM BRAIN -INSPIRED INTELLIGENCE TO EVOLUTIONARY ,COLLABORATIVE ,AND SAFE SYSTEMS Bang Liu2,3,20→†, Xinfeng Li4→, Jiayi Zhang1,10→, Jinlin Wang1→, Tanjin He5→, Sirui Hong1→, Hongzhang Liu6→,Shaokun Zhang7→,Kaitao Song8→,Kunlun

The endowment of AI with reasoning capabilities and some degree of agency is widely viewed as a path toward more capable and generalizable sys- tems. Our position is that the current development of agentic AI requires a more holistic, systems- theoretic perspective in order to fully understand their capabilities and mitigate any emergent risks. The

—This review critically distinguishes between AI Agents and Agentic AI, offering a structured conceptual tax- onomy, application mapping, and challenge analysis to clarify their divergent design philosophies and capabilities. We begin by outlining the search strategy and foundational definitions, characterizing AI Agents as modular systems driven by LLMs

arXiv:2501.09223v1 [cs.CL] 16 Jan 2025Foundations of Large Language Models Tong Xiao and Jingbo Zhu January 17, 2025 NLP Lab, Northeastern University & NiuTrans Research Copyright © 2021-2025 Tong Xiao and Jingbo Zhu NATURAL LANGUAGE PROCESSING LAB, NORTHEASTERN UNIVERSITY & NIUTRANS RESEARCH Licensed

DECODING TRUST : A Comprehensive Assessment of Trustworthiness in GPT Models Boxin Wang1→, Weixin Chen1→, Hengzhi Pei1→, Chulin Xie1→, Mintong Kang1→, Chenhui Zhang1→, Chejian Xu1, Zidi Xiong1, Ritik Dutta1, Rylan Schaeffer2, Sang T. Truong2, Simran Arora2, Mantas Mazeika1, Dan Hendrycks3,4, Zinan Lin5, Yu

ing the topic since 2008. High-Flyer is reported to have 113 employees and $59 million in revenues while manag - ing $8 billion in assets.11,c During 2019–2022, High-Flyer spent approximately $139 million and stockpiled 10,000 Nvidia A100 GPUs to build a supercomputer. The A100 is lower in performance than the datacenter-standard Nvidia H100 (the

Nature | Vol 645 | 18 September 2025 | 633 ArticleDeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning General reasoning represents a long-standing and formidable challenge in artificial intelligence (AI). Recent breakthroughs, exemplified by large language models (LLMs)1,2 and chain-of-thought

We introduce our first-generation reasoning models, DeepSeek-R1-Zero and DeepSeek-R1. DeepSeek-R1-Zero, a model trained via large-scale reinforcement learning (RL) without super- vised fine-tuning (SFT) as a preliminary step, demonstrates remarkable reasoning capabilities. Through RL, DeepSeek-R1-Zero naturally

As the capabilities of code large language models (LLMs) continue to expand, their applica- tions across diverse code intelligence domains are rapidly increasing. However, most existing datasets only evaluate limited application domains. To address this gap, we have developed a comprehensive code evaluation dataset FullStack Bench1,2focusing on full-

—Large Language Models (LLMs) have transformed the natural language processing landscape and brought to life diverse applications. Pretraining on vast web-scale data has laid the foundation for these models, yet the research community is now increasingly shifting focus toward post-training techniques to achieve further breakthroughs. While

Published as a conference paper at COLM 2025 Self-Steering Language Models Gabriel Grand1∗Joshua B. Tenenbaum1Vikash K. Mansinghka1 Alexander K. Lew1,2Jacob Andreas1 1Massachusetts Institute of Technology2Yale University {gg, jbt, vkm, jda}@mit.edu alexander.lew@yale.edu /githubgithub.com/gabegrand/self-

Rethinking Memory in AI: Taxonomy, Operations, Topics, and Future Directions Yiming Du1,2∗, Wenyu Huang2∗, Danna Zheng2∗, Zhaowei Wang3, Sebastien Montella4,Mirella Lapata2,Kam-Fai Wong1,Jeff Z. Pan2,4 1The Chinese University of Hong Kong2The University of Edinburgh3HKUST 4Poisson Lab, CSI, Huawei UK R&D Ltd.

The large language models, as predicted by scaling law forecasts, have made groundbreak - ing progress in many fields, particularly in natural language generation tasks, where they have approached or even surpassed human levels. However, the unprecedented scale of their parameters brings significant computational and storage costs. These large

MapReduce: Simplied Data Processing onLargeClusters JeffreyDean andSanjay Ghema wat jeff@google.com, sanjay@google.com Google,Inc. Abstract MapReduce isaprogramming model andanassoci- ated implementation forprocessing andgenerating large data sets. Users specify amap function thatprocesses a

The Entity-Relationship Model-Toward a Unified View of Data PETER PIN-SHAN CHEN Massachusetts Institute of Technology A data model, called the entity-relationship model, is proposed. This model incorporates some of the important semantic information about the real world. A special diagrammatic technique is introduced as a

QueryEvaluation Techniques forLargeDatabases GOETZ GRAEFE Portland StateUniversity, Computer Science Department, P.O.Box751, Portland, Oregon 97207-0751 Database management systemswillcontinue tomanagelargedatavolumes. Thus, efficientalgorithms foraccessing andmanipulating largesetsandsequences

of Query Optimization in Relational Systems Surajit Chaudhuri Microsoft Research One Microsoft Way Redmond, WA 98052 +1-(425)-703-1938 surajitc@microsoft.com

Classes of small-world networks L. A. N. Amaral*, A. Scala, M. Barthe ´le´my†, and H. E. Stanley Center for Polymer Studies and Department of Physics, Boston University, Boston, MA 02215 Communicated by Herman Z. Cummins, City College of the City University of New York, New York, NY, July 13, 2000 (received for review April

Unlike clearly defined 'real-world' net- works1, social networks tend to be subjective to some extent2,3because the perception of what constitutes a sociallink may differ between individuals. Oneunambiguous type of connection, however,is sexual contact, and here we analyse thesexual behaviour of a random sample

DataCleaning: ProblemsandCurrentApproaches Erhard Rahm /A3HongHai Do UniversityofLeipzig,Germany http://dbs.uni-leipzig.de Abstract We classify data quality problems that are addressed by data cleaning and provide an overview of the main solution approaches. Data cleaning is especially required when integrating heterogeneous

We have designed and implemented the Google File Sys- tem, a scalable distributed file system for large distributeddata-intensiveapplications. It providesfault tolerance whilerunningoninexpensivecommodityhardware, anditdelivers high aggregate performance to a large number

This paper presents the design of a read-optimized relational DBMS that contrasts sharply with most current systems, which are write-optimized. Among the many differences in its design are: storage of data by column rather than by row, careful coding and packing of objects into storage including main memory during query

Institute of Mathematical Statistics is collaborating with JSTOR to digitize, preserve, and extend access to The Annals of Mathematical Statistics. www.jstor.org®

—Often, in the real world, entities have two or more representations in databases. Duplicate records do not share a common key and/or they contain errors that make duplicate matching a difficult task. Errors are introduced as the result of transcription errors,incomplete information, lack of standard formats, or any combination of these factors. In this

MapReduce: Simplified Data Processing on Large Clusters by Jeffrey Dean and Sanjay Ghemawat 1 Introduction Prior to our development of MapReduce, the authors and many others at Google implemented hundreds of special-purpose computations thatprocess large amounts of raw data, such as crawled

On the Theory of Scales of Measurement S. S. Stevens Science, New Series, Vol. 103, No. 2684. (Jun. 7, 1946), pp. 677-680. Stable URL: http://links.jstor.org/sici?sici=0036-8075%2819460607%293%3A103%3A2684%3C677%3AOTTOSO%3E2.0.CO%3B2-G Scienceis currently published

We Need Both Exploratory and Confirmatory Author(s): John W. Tukey Source: The American Statistician, Vol. 34, No. 1 (Feb., 1980), pp. 23-25 Published by: American Statistical Association Stable URL: http://www.jstor.org/stable/2682991 Accessed: 21/12/2009 06:38 Your use of the JSTOR archive indicates your

arXiv:cond-mat/0007214v1 [cond-mat.stat-mech] 12 Jul 2000The structure of scientific collaboration networks M. E. J. Newman Santa Fe Institute, 1399 Hyde Park Road, Santa Fe, NM 87501 We investigate the structure of scientific collaboration ne tworks. We consider two scientists to be connected if they have authored a paper

Beyond Accuracy: What Data Quality Means to Data Consumers Author(s): Richard Y. Wang and Diane M. Strong Source: Journal of Management Information Systems, Vol. 12, No. 4 (Spring, 1996), pp. 5-33 Published by: M.E. Sharpe, Inc. Stable URL: http://www.jstor.org/stable/40398176 . Accessed: 26/10/2013 17:54 Your use of

Dremel is a scalable, interactive ad-hoc query system for analy- sis of read-only nested data. By combining multi-level execution trees and columnar data layout, it is capable of running aggrega- tion queries over trillion-row tables in seconds. The system scales to thousands of CPUs and petabytes of data, and has thousands of

Spark: Cluster Computing with Working Sets Matei Zaharia, Mosharaf Chowdhury, Michael J. Franklin, Scott Shenker, Ion Stoica University of California, Berkeley Abstract MapReduce and its variants have been highly successful in implementing large-scale data-intensive applications on commodity clusters. However, most of

FeBRuARY 2012 | VoL. 55 | No. 2 | coMMunica Tions of The ac M 81in performing them for terabyte or larger datasets (increasingly common across scientific disciplines) are quite different from those that applied when data volumes were measured in kilobytes. The result is a computa - tional crisis in many laboratories and a

Make Data Work strataconf.com Presented by O'Reilly and Cloudera, Strata + Hadoop World is where cutting-edge data science and new business fundamentals intersect— and merge. n Learn business applications of data technologies n Develop new skills through trainings and in-depth tutorials n Connect with an international

Storm has long served as the main platform for real-time analytics at Twitter. However, as the scale of data being processed in real- time at Twitter has increased, al ong with an increase in the diversity and the number of use cas es, many limitations of Storm have become apparent. We need a system that scales better, has better debug-ability, has

73Chapter 5CHAPTER FIVE Information Platforms and the Rise of the Data Scientist Jeff Hammerbacher Libraries and Brains AT THE AGE OF 17, I WAS FIRED FROM MY JOB AS A CASHIER AT SCOTT 'S GROCERY STORE IN FORT Wayne, Indiana. With only two months remaining before my freshman year of college, I saw in

8/20/2015 Software as a Service for Data Scientists | February 2012 | Communications of the ACM http://cacm.acm.org/magazines/2012/2/145400-software-as-a-service-for-data-scientists/fulltext 1/8CONTRIBUTED ARTICLES Software as a Service for Data Scientists By Bryce Allen, John Bresnahan, Lisa Childers, Ian Foster, Gopi

ion | LinkedIn Engineering https://engineering.linkedin.com/distributed-systems/log-what-every-software-engineer-should-know-about-real-time-datas-unifying 1/19HomeProjectsTechnologyTeam BlogTech TalksJobs Engineering Follow @LinkedInEngLinkedIn.com The Log: What every software engineer should know about

8/20/2015 The Netflix Tech Blog: Hadoop Platform as a Service in the Cloud http://techblog.netflix.com/2013/01/hadoop-platform-as-service-in-cloud.html 1/7 Thursday, January 10, 2013 Hadoop Platform as a Service in the Cloud by Sriram Krishnan and Eva Tse, Cloud Big Data Platform Hadoop

Kafka: a Distributed Messaging System for Log Processing Jay Kreps LinkedIn Corp. jkreps@linkedin.comNeha Narkhede LinkedIn Corp. nnarkhede@linkedin.comJun Rao LinkedIn Corp. jrao@linkedin.com ABSTRACT Log processing has become a critical component of the data pipeline for consumer internet companies. We

Dimensions for Characterizing Analytics Data-Processing Solutions Jari Koister August 2015 Introduction Data analytics and business intelligence have been of interest for a long time, and their needs have been addressed by various technologies based on relational technologies and cubes. Columnar databases and in-memory

The software processes that facilitate the original loading and the periodic refreshment of the data warehouse contents are commonly known as Extraction-Transformation-Loading (ETL) processes. The intention of this survey is to present the research work in the field of ETL technology in a structured way. To this end, we organize the coverage of the field

92 COMMUNICATIONS OF THE ACM | FEBRUARY 2016 | VOL. 59 | NO. 2review articles A GROUP OF database researchers meets periodically to discuss the state of the field and its key directions going forward. Past meetings were held in 1989,6 1990,11 1995,12 1996,10 1998,7 2003,1 and 2008.2 Continuing this tradition, 28 database researchers

Formal Mathematical Reasoning: A New Frontier in AI Kaiyu Yang1, Gabriel Poesia2, Jingxuan He3, Wenda Li4, Kristin Lauter1, Swarat Chaudhuri5, Dawn Song3 1Meta FAIR,2Stanford University,3UC Berkeley,4University of Edinburgh,5UT Austin Abstract AI for Mathematics (AI4Math) is not only intriguing intellectually but also crucial

ATaxonomy ofComputer Program Security Flaws CARLE.LANDWEHR, ALANR.BULL,JOHNP.MCDERMOTT, ANDWILLIAM S.CHOI Informatmn Technology Division, NavalResearch Laboratory, Washington, D.C.20375-5337 Anorganized record ofactual flawscanbeusefultocomputer system designers, programmers, analysts, administrators,

today are by and large unaware of the security problems they can (un-knowingly) introduce into code, ataxonomy of coding errors shouldprovide a real tangible benefit to thesoftware security community. This approach represents a strik- ing alternative to taxonomies of at-tack patterns 1or simple-minded collections of specific vulnerabilities(such as M

Machinelearningsystemsofferunparalledflexibilityindeal- ingwithevolvinginputinavarietyofapplications,suchas intrusiondetectionsystemsandspame-mailfiltering. How- ever,machinelearningalgorithmsthemselvescanbeatarget of attack by a malicious adversary. This paper provides a frameworkforansweringthequestion,"Canmachinelearn- ing be

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Machine learning's ability to rapidly evolve to changing and complex situations has helped it become a fundamental tool for computer security. That adaptability is alsoa vulnerability: attackers can exploit machine learning systems. We present a taxonomy identifying and analyzing attacks against machine learning systems. We show how these classes

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In this paper (expanded from an invited talk at AISEC 2010), we discuss an emerging field of study: adversarial machine learning—the study of effective machine learning tech- niques against an adversarial opponent. In this paper, we: give a taxonomy for classifying attacks against online ma- chine learning algorithms; discuss application-specific factors that limit an adversary's capabilities in

In this paper (expanded from an invited talk at AISEC 2010), we discuss an emerging field of study: adversarial machine learning—the study of effective machine learning tech- niques against an adversarial opponent. In this paper, we: give a taxonomy for classifying attacks against online ma- chine learning algorithms; discuss application-specific factors that limit an adversary's capabilities in

Popular Internet sites are under attack all the time from phishers, fraudsters, and spammers. They aim to steal user information and expose users to unwanted spam. The attackers have vast resources at their disposal. They are well-funded, with full-time skilled labor,control over compromised and infected accounts, and access to global botnets. Protecting our users is a challenging adversarial com

Adaptive biometric systems update clients' templates during operation to account for natural changes over time (e.g., aging of biometric templates). Recently, it has been shown that this update can be exploited by an attacker to compromise the clients' templates: by presenting a proper sequence of fake biometric traits to the sensor, the attacker may eventually impersonate the targeted clients wit

Intriguing properties of neural networks Christian Szegedy Google Inc.Wojciech Zaremba New York UniversityIlya Sutskever Google Inc.Joan Bruna New York University Dumitru Erhan Google Inc.Ian Goodfellow University of MontrealRob Fergus New York University Facebook Inc. Abstract Deep neural networks are highly expressive models that have recently achieved state of the art performance on speech and

Machine-learning (ML) algorithms are increasingly utilized in privacy-sensitive applications such as predicting lifestyle choices, making medical diagnoses, and facial recognition. In a model inversion attack, recently introduced in a case study of linear classifiers in personalized medicine by Fredrikson et al. [13], adversarial access to an ML model is abused to learn sensitive genomic informatio

Pattern Recognitionand ApplicationsLab Universityof Cagliari, ItalyDepartment of Electrical and Electronic Engineering ICCV'17 –Half-day Tutorial onAdversarial Pattern RecognitionBattista Biggioand Fabio Roli ICCV '17, Venice, Italy, October 22, 2017http://pralab.diee.unica.it/en/wild-patterns http://pralab.diee.unica.it Deep face recognition… 3DeepNeuralNetwork[O.M. Parkhi et al., Deep face recogn

Published as a conference paper at ICLR 2015 EXPLAINING AND HARNESSING ADVERSARIAL EXAMPLES Ian J. Goodfellow, Jonathon Shlens & Christian Szegedy Google Inc., Mountain View, CA {goodfellow,shlens,szegedy }@google.com ABSTRACT Several machine learning models, including neural networks, consistently mis- classify adversarial examples—inputs formed by applying small but intentionally worst-case per

Scalable Optimization of Randomized Operational Decisions in Adversarial Classification Settings Bo Li and Yevgeniy Vorobeychik Electrical Engineering and Computer Science Vanderbilt University Nashville, TN {bo.li.2,yevgeniy.vorobeychik }@vanderbilt.edu Abstract When learning, such as classification, is used in adversarial settings, such as intrusion detection, intelligent adversaries will attempt

The Limitations of Deep Learning in Adversarial Settings Nicolas Papernot∗, Patrick McDaniel∗, Somesh Jha†, Matt Fredrikson‡, Z. Berkay Celik∗, Ananthram Swami§ ∗Department of Computer Science and Engineering, Penn State University †Computer Sciences Department, University of Wisconsin-Madison ‡School of Computer

Forecasting models play a key role in money-making ven- tures in many different markets. Such models are often trained on data from various sources, some of which may be untrustworthy. An actor in a given market may be incen- tivised to drive predictions in a certain direction to their own benefit. Prior analyses of intelligent adversaries in a machine learning context have focused on regression a

Machine learning is enabling a myriad innovations, including new algorithms for cancer diagnosis and self-driving cars. The broad use of machine learning makes it important to understand the extent to which machine-learning algorithms are subject to attack, particularly when used in applications where physical security or safety is at risk. In this paper, we fo cus on facial biometric systems, whic

Adversarial AI Yevgeniy Vorobeychik Electrical Engineering and Computer Science Vanderbilt University yevgeniy.vorobeychik@vanderbilt.edu Abstract In recent years AI research has had an increasing role in models and algorithms for security prob- lems. Game theoretic models of security, and Stackelberg security games in particular, have received special attention, in part because these models a

SoK: Towards the Science of Security and Privacy in Machine Learning Nicolas Papernot∗, Patrick McDaniel∗, Arunesh Sinha†, and Michael Wellman† ∗Pennsylvania State University †University of Michigan {ngp5056,mcdaniel }@cse.psu.edu, {arunesh,wellman }@umich.edu Abstract —Advances in machine learning (ML) in recent years have enabled a dizzying array of applications such as data analytics, autonomou

This paper is included in the Proceedings of the 25th USENIX Security Symposium August 10–12, 2016 • Austin, TX ISBN 978-1-931971-32-4 Open access to the Proceedings of the 25th USENIX Security Symposium is sponsored by USENIX Stealing Machine Learning Models via Prediction APIs Florian Tramèr, École Polytechnique Fédérale de Lausanne (EPFL) Fan Zhang, Cornell University Ari Juels, Cornell Tech

This paper is included in the Proceedings of the 25th USENIX Security Symposium August 10–12, 2016 • Austin, TX ISBN 978-1-931971-32-4 Open access to the Proceedings of the 25th USENIX Security Symposium is sponsored by USENIX Stealing Machine Learning Models via Prediction APIs Florian Tramèr, École Polytechnique Fédérale de Lausanne (EPFL) Fan Zhang, Cornell University Ari Juels, Cornell Tech

Many machine learning models are vulnerable to adversarial examples : inputs that are specially crafted to cause a machine learning model to produce an incorrect output. Ad- versarial examples that affect one model often affect another model, even if the two models have different architectures or were trained on different training sets, so long as both models were trained to perform the same task.

123Machine Learning ISSN 0885-6125Volume 106Number 7 Mach Learn (2017) 106:1039-1082DOI 10.1007/s10994-017-5633-9Optimal classification treesDimitris Bertsimas & Jack Dunn

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/317296994Adversarial learning in credit card fraud detectionConference Paper · April 2017DOI: 10.1109/SIEDS.2017.7937699CITATIONS8READS7026 authors, including: Some of the authors of this

Certified Defenses for Data Poisoning Attacks Jacob Steinhardt∗ Stanford University jsteinha@stanford.eduPang Wei Koh∗ Stanford University pangwei@cs.stanford.eduPercy Liang Stanford University pliang@cs.stanford.edu Abstract Machine learning systems trained on user-provided data are susceptible to data poisoning attacks, whereby malicious

Certified Defenses for Data Poisoning Attacks Jacob Steinhardt∗ Stanford University jsteinha@stanford.eduPang Wei Koh∗ Stanford University pangwei@cs.stanford.eduPercy Liang Stanford University pliang@cs.stanford.edu Abstract Machine learning systems trained on user-provided data are susceptible to data poisoning attacks, whereby malicious

Imperial College London Department of Computing BEng Computing Final Year Individual Project Report Can you Poison a Machine Learning Algorithm? Author: Vasin WongrassameeSupervisor: Dr. Luis Muñoz-González June 19, 2017 Acknowledgements I would like to thank my supervisor, Dr. Luis Muñoz-González, for his expert advice and

Membership Inference Attacks Against Machine Learning Models Reza Shokri Cornell Tech shokri@cornell.eduMarco Stronati← INRIA marco@stronati.orgCongzheng Song Cornell cs2296@cornell.eduVitaly Shmatikov Cornell Tech shmat@cs.cornell.edu Abstract —We quantitatively investigate how machine learning models leak information about

Machine learning (ML) models, e.g., deep neural networks (DNNs), are vulnerable to adversarial examples: malicious inputs modified to yield erroneous model outputs, while ap- pearing unmodified to human observers. Potential attacks include having malicious content like malware identified as legitimate or controlling vehicle behavior. Yet, all existing

Operationalizing machine learning based security detections is ex- tremely challenging, especially in a continuously evolving cloud environment. Conventional anomaly detection does not produce satisfactory results for analysts that are investigating security in- cidents in the cloud. Model evaluation alone presents its own set of problems due to a lack of

Machine learning based system are increasingly being used for sensitive tasks such as security surveillance , guiding autonomous vehicle, taking investment decisions, detecting and blocking network intrusion and malware etc. However, recent research has shown that machine learning models ar e venerable to attacks by adversaries at all phases of

Security Risks in Deep Learning Implementations Qixue Xiao1, Kang Li2, Deyue Zhang1, Weilin Xu3 1Qihoo 360 Security Research Lab2University of Georgia3University of Virginia Abstract —Advance in deep learning algorithms overshadows their security risk in software implementations. This paper dis- closes a set of

Towards Evaluating the Robustness of Neural Networks Nicholas Carlini David Wagner University of California, Berkeley ABSTRACT Neural networks provide state-of-the-art results for most machine learning tasks. Unfortunately, neural networks are vulnerable to adversarial examples: given an input xand any target classification t, it is possible to

Towards Evaluating the Robustness of Neural Networks Nicholas Carlini David Wagner University of California, Berkeley ABSTRACT Neural networks provide state-of-the-art results for most machine learning tasks. Unfortunately, neural networks are vulnerable to adversarial examples: given an input xand any target classification t, it is possible to

Y ou Can Hear But Y ou Cannot Steal: Defending against V oice Impersonation Attacks on Smartphones Si Chen†‡, Kui Ren†, Sixu Piao†, Cong Wang¶, Qian Wang§, Jian Weng♮,L uS u†, Aziz Mohaisen† †Department of Computer Science and Engineering, University at Buffalo, SUNY ‡Department of Computer Science, West Chester University

Published as a conference paper at ICLR 2018 GENERATING NATURAL ADVERSARIAL EXAMPLES Zhengli Zhao University of California Irvine, CA 92697, USA zhengliz@uci.eduDheeru Dua University of California Irvine, CA 92697, USA ddua@uci.eduSameer Singh University of California Irvine, CA 92697, USA sameer@uci.edu ABSTRACT Due to

56 COMMUNICATIONS OF THE ACM | JULY 2018 | VOL. 61 | NO. 7contributed articles SHEEP AND WOLF ILLUSTRATIONS BY ARN0 MACHINE LEARNING HAS advanced radically over the past 10 years, and machine learning algorithms now achieve human-level performance or better on a number of tasks, including face recognition,31

56 COMMUNICATIONS OF THE ACM | JULY 2018 | VOL. 61 | NO. 7contributed articles SHEEP AND WOLF ILLUSTRATIONS BY ARN0 MACHINE LEARNING HAS advanced radically over the past 10 years, and machine learning algorithms now achieve human-level performance or better on a number of tasks, including face recognition,31

Adversarial examplesrevealedtheweaknessofmachinelearningtechniquesintermsofrobustness,whichmoreover inspired adversariestomakeuseoftheweaknesstoattacksystemsemployingmachinelearning.Existingresearches coveredthemethodologiesofadversarialexamplegeneration,therootreasonoftheexistenceofadversarial

Adversarial Attacks Against Medical Deep Learning Systems Samuel G. Finlayson Department of Systems Biology Harvard Medical School Boston, MA 02130 samuel_finlayson@hms.harvard.eduHyung Won Chung Massachusetts Institute of Technology Cambridge, MA 02139 hwc@mit.edu Isaac S. Kohane Department of

x Adversarial Attacks and Defences: A Survey ANIRBAN CHAKRABORTY∗,Indian Institute of Technology, Kharagpur, India MANAAR ALAM, Indian Institute of Technology, Kharagpur, India VISHAL DEY, The Ohio State University, Columbus, United States ANUPAM CHATTOPADHYAY, Nanyang Technological University, Singapore DEBDEEP

x Adversarial Attacks and Defences: A Survey ANIRBAN CHAKRABORTY∗,Indian Institute of Technology, Kharagpur, India MANAAR ALAM, Indian Institute of Technology, Kharagpur, India VISHAL DEY, The Ohio State University, Columbus, United States ANUPAM CHATTOPADHYAY, Nanyang Technological University, Singapore DEBDEEP MUKHOPADHYAY, Indian Institute of Technology, Kharagpur, India Deep learning has emerg

Deep learning models for graphs have achieved strong performance for the task of node classification. Despite their proliferation, cur- rently there is no study of their robustness to adversarial attacks. Yet, in domains where they are likely to be used, e.g. the web, adver- saries are common. Can deep learning models for graphs be easily fooled? In this work, we introduce the first study of adver

Machine learning (ML) systems have been increasingly applied in web security applications such as spammer detection, malware detection and fraud detection. These applications have an intrin- sic adversarial nature where intelligent attackers can adaptively change their behaviors to avoid being detected by the deployed detectors. Existing efforts against adversaries are usually limited by the type

Audio Adversarial Examples: Targeted Attacks on Speech-to-Text Nicholas Carlini David Wagner University of California, Berkeley Abstract —We construct targeted audio adversarial examples on automatic speech recognition. Given any audio waveform, we can produce another that is over 99.9% similar, but transcribes as any phrase we choose (recognizing up to 50 characters per second of audio). We apply

Detecting Poisoning Attacks on Machine Learning in IoT Environments Nathalie Baracaldo, Bryant Chen, Heiko Ludwig, Amir Safavi, Rui Zhang IBM Almaden Research Center San Jose, CA, United States baracald@us.ibm.com, bryant.chen@ibm.com, hludwig@us.ibm.com, amir.safavi@ibm.com, ruiz@us.ibm.com Abstract —Machine Learning (ML) plays an increasing role in Internet of Things (IoT), both in the Cloud and

In a world that relies increasingly on large amounts of data and on powerful Machine Learning (ML) models, the veracity of decisions made by these systems is essential. Adversarial samples are inputs that have been perturbed to mislead the in- terpretation of the ML and are a dangerous vulnerability. Our research takes a first

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Manipulating Machine Learning: Poisoning Attacks and Countermeasures for Regression Learning Matthew Jagielski∗, Alina Oprea∗, Battista Biggio†‡, Chang Liu§, Cristina Nita-Rotaru∗, and Bo Li§ ∗Northeastern University, Boston, MA†University of Cagliari, Italy‡Pluribus One, Italy§UC Berkeley, Berkeley, CA Abstract —As machine learning becomes widely used for auto- mated decisions, attackers have str

Data poisoning is a type of adversarial attack on machine learning models wherein the attacker adds examples to the training set to manipulate the behavior of the model at test time. This paper explores a broad class of poisoning attacks on neural nets. The proposed attacks use “clean-labels”; they don’t require the attacker to have any control over the labeling of training data. They are also stealthy

Data poisoning is an attack on machine learning models wherein the attacker adds examples to the training set to manipulate the behavior of the model at test time. This paper explores poisoning attacks on neural nets. The proposed attacks use “clean-labels”; they don’t require the attacker to have any control over the labeling of training data. They are also targeted; they control the behavior of

Published as a conference paper at ICLR 2018 GENERATING NATURAL ADVERSARIAL EXAMPLES Zhengli Zhao University of California Irvine, CA 92697, USA zhengliz@uci.eduDheeru Dua University of California Irvine, CA 92697, USA ddua@uci.eduSameer Singh University of California Irvine, CA 92697, USA sameer@uci.edu ABSTRACT Due to their complex nature, it is hard to characterize the ways in which machine lea

Published as a conference paper at ICLR 2018 EVALUATING THE ROBUSTNESS OF NEURAL NET- WORKS : ANEXTREME VALUE THEORY APPROACH Tsui-Wei Weng1∗, Huan Zhang2∗, Pin-Yu Chen3, Jinfeng Yi4, Dong Su3, Yupeng Gao3, Cho-Jui Hsieh2, Luca Daniel1 1Massachusetts Institute of Technology, Cambridge, MA 02139 2University of California, Davis, CA 95616 3IBM Research AI, Yorktown Heights, NY 10598 4Tencent AI Lab

Published as a conference paper at ICLR 2018 SCALABLE PRIVATE LEARNING WITH PATE Nicolas Papernot∗ Pennsylvania State University ngp5056@cse.psu.eduShuang Song∗ University of California San Diego shs037@eng.ucsd.edu Ilya Mironov, Ananth Raghunathan, Kunal Talwar & Úlfar Erlingsson Google Brain {mironov,ananthr,kunal,ulfar}@google.com ABSTRACT The rapid adoption of machine learning has increased

Machine learning is one of the most prevailing techniques in computer science, and it has been widely applied in image processing, natural language processing, pattern recognition, cybersecurity, and other fields. Regardless of successful applications of machine learning algorithms in many scenarios, e.g., facial recognition, malware detection, automatic driving, and intrusion detection, these algo

—Advances in machine learning (ML) in recent years have enabled a dizzying array of applications such as data analytics, autonomous systems, and security diagnostics. ML is now pervasive—new systems and models are being deployed in every domain imaginable, leading to widespread deployment of software based inference and decision making. There is growing recognition that ML exposes new vulnerabilit

In this paper, a novel method using 3D Convolutional Neural Network (3D-CNN) architecture has been proposed for speaker verification in the text-independent setting. One of the main challenges is the creation of the speaker models. Most of the previously-reported approaches create speaker models based on averaging the extracted features from utterances of the speaker, which is known as the d-vector

The Secret Sharer: Measuring Unintended Neural Network Memorization & Extracting Secrets Nicholas Carlini University of California, BerkeleyChang Liu University of California, Berkeley Jernej Kos National University of Singapore´Ulfar Erlingsson Google BrainDawn Song University of California, Berkeley Abstract Machine learning models based on neural networks and deep learning are being rapidly adop

Recent studies show that the state-of-the-art deep neural networks (DNNs) are vulnerable to adversarial examples, resulting from small-magnitude perturbations added to the input. Given that that emerging physical systems are us- ing DNNs in safety-critical situations, adversarial examples could mislead these systems and

This paper is included in the Proceedings of the 27th USENIX Security Symposium. August 15–17, 2018 • Baltimore, MD, USA ISBN 978-1-931971-46-1 Open access to the Proceedings of the 27th USENIX Security Symposium is sponsored by USENIX.Skill Squatting Attacks on Amazon Alexa Deepak Kumar, Riccardo Paccagnella, Paul Murley, Eric Hennenfent, Joshua Mason, Adam Bates, and Michael Bailey, University

Deep learning is at the heart of the current rise of artificial intelligence. In the field of computer vision, it has become the workhorse for applications ranging from self-driving cars to surveillance and security. Whereas, deep neural networks have demonstrated phenomenal success (often beyond human capabilities) in solving

Learning-based pattern classifiers, including deep networks, have shown impressive performance in several application domains, ranging from computer vision to cybersecurity. However, it has also been shown that adversarial input perturbations carefully crafted either at training or at test time can easily subvert their predictions. The vulnerability of machine learning to such wild patterns (also r

Adversarial examplesrevealedtheweaknessofmachinelearningtechniquesintermsofrobustness,whichmoreover inspired adversariestomakeuseoftheweaknesstoattacksystemsemployingmachinelearning.Existingresearches coveredthemethodologiesofadversarialexamplegeneration,therootreasonoftheexistenceofadversarial examples and deemed defense schemes. However, practical attacks against real-world systems, mostly recent mai

Draft NIST IR 8269 1 2 A Taxonomy and Terminology of 3 Adversarial Machine Learning 4 5 Elham Tabassi 6 Kevin J. Burns 7 Michael Hadjimichael 8 Andres D. Molina -Markham 9 Julian T. Sexton 10 11 12 13 14 This publication is available free of charge from: 15 https://doi.org/10.6028/NIST.IR.8269 -draft 16 17

Can machine learning models be easily fooled? Despite the recent surge of interest in learned adversarial attacks in other domains, in the context of recommendation systems this question has mainly been answered using hand-engineered fake user profles. This paper attempts to reduce this gap. We provide a formulation for learning to attack a recommender as a repeated general-sum game between two pl

applied sciences ReviewReview of Artificial Intelligence Adversarial Attackand Defense TechnologiesShilin Qiu†, Qihe Liu *,†, Shijie Zhou†and Chunjiang Wu†The School of Information and Software Enginerring, University of Electronic Science and Technology of China,Chengdu 610054, China; qiushilin@std.uestc.edu.cn (S.Q.); sjzhou@uestc.edu.cn (S.Z.) ychunjiangwu@126.com (C.W.) *Correspondence: qihe.liu@u

It is well-known that there is no safety without security. That being said, a sound investigation of security breaches on Ma- chine Learning (ML) is a prerequisite for any safety concerns. Since attacks on ML systems and their impact on the security goals threaten the safety of an ML system, we discuss the im- pact attacks have on the ML models' security goals, which are rarely considered in publi

..... 1 Research Target
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Evading Real-Time Person Detectors by Adversarial T-shirt Kaidi Xu1Gaoyuan Zhang2Sijia Liu2Quanfu Fan2Mengshu Sun1 Hongge Chen3Pin-Yu Chen2Yanzhi Wang1Xue Lin1 1Northeastern University, USA 2MIT-IBM Watson AI Lab, IBM Research, USA 3Massachusetts Institute of Technology, USA October 25, 2019 Abstract It is known that deep neural networks (DNNs) could be vulnerable to adversarial attacks. The so-ca

Machine learning systems have had enormous success in a wide range of fields from computer vision, natural language process- ing, and anomaly detection. However, such systems are vulnerable to attackers who can cause deliberate misclassification by intro- ducing small perturbations. With machine learning systems being proposed for cyber attack detection, such attackers are cause for serious concern.

Recent work has demonstrated that deep neural networks are vulnerable to adversarial examples—inputs that are almost indistinguishable from natural data and yet classified incor- rectly by the network. In fact, some of the latest findings suggest that the existence of adversarial attacks may be an inherent weakness of deep

ADVERSARIAL ATTACKS ON MACHINE LEARNING SYSTEMS FOR HIGH-FREQUENCY TRADING Micah Goldblum∗ University of Maryland goldblum@umd.eduAvi Schwarzschild∗ University of Maryland Ankit Patel Rice University, Baylor College of MedicineTom Goldstein University of Maryland ABSTRACT Algorithmic trading systems are often completely automated, and deep learning is increasingly receiving attention in this domai

INVITED: AI Utopia or Dystopia - On Securing AI Platforms Ghada Dessouky ghada.dessouky@trust.tu-darmstadt.de Technical University of Darmstadt, GermanyPatrick Jauernig patrick.jauernig@trust.tu-darmstadt.de Technical University of Darmstadt, GermanyNele Mentens nele.mentens@kuleuven.be KU Leuven, Belgium Ahmad-Reza Sadeghi ahmad.sadeghi@trust.tu-darmstadt.de Technical University of Darmstadt, Ger

We study data poisoning attacks in the online learning setting, where training data arrive sequentially, and the attacker is eavesdropping the data stream and has the ability to contaminate the current data point to affect the online learning process. We formulate the optimal online attack problem as a stochastic optimal control problem, and provide a systematic solution using tools from model pre

AI for Security and Security for AI Elisa Bertino bertino@purdue.edu Purdue UniversityMurat Kantarcioglu muratk@utdallas.edu The University of Texas at DallasCuneyt Gurcan Akcora Cuneyt.Akcora@umanitoba.ca University of Manitoba, Canada Sagar Samtani ssamtani@iu.edu Indiana UniversitySudip Mittal (Moderator) mittals@uncw.edu University of North Carolina WilmingtonMeenak Gupta (Moderator) mgupta@tr

Many powerful computing technologies rely on implicit and ex- plicit data contributions from the public. This dependency suggests a potential source of leverage for the public in its relationship with technology companies: by reducing, stopping, redirecting, or oth- erwise manipulating data contributions, the public can reduce the effectiveness of many lucrative technologies. In this paper, we ex

Dataset Security for Machine Learning: Data Poisoning, Backdoor Attacks, and Defenses Micah Goldblum *1, Dimitris Tsipras2, Chulin Xie3, Xinyun Chen4, Avi Schwarzschild1, Dawn Song4, Aleksander M ˛ adry2, Bo Li3, and Tom Goldstein†1 1University of Maryland 2Massachussetts Institute of Technology 3University of Illinois Urbana-Champaign

JUNE 2021 | VOL. 64 | NO. 6 | COMMUNICATIONS OF THE ACM 15newsIMAGE BY ANDRIJ BORYS ASSOCIATES, USING SHUTTERSTOCKtend to assume that they see the same way we do," said Draper. "That's not a good assumption." Troublingly, this vulnerability likely extends to other applications of machine learning, where the results are

The majority of machine learning methodologies operate with the assumption that their environment is benign. However, this assumption does not always hold, as it is often advantageous to adversaries to maliciously modify the train- ing (poisoning attacks) or test data (evasion attacks). Such attacks can be catas- trophic given the growth and the

Given the computational cost and technical expertise required to train machine learning models, users may delegate the task of learning to a service provider. Delegation of learning has clear benefits, and at the same time raises serious concerns of trust . This work studies possible abuses of power by untrusted learners. We show how a malicious learner

Coding errors are a critical problem in software security. This Research Report shows the common types of coding errors organized into a simple taxonomy to help developers and security practitioners recognize the categories of problems that lead to vulnerabilities and help them identify existing errors as they build software. The information

Large language models (LLMs) are being deployed in many domains of our lives ranging from browsing, to voice assistants, to coding assistance tools, and have potential for vast societal impacts.[1,2,3,4,5,6,7] This system card analyzes GPT-4, the latest LLM in the GPT family of models.[8,9,10] First, we highlight safety

Generative

Learning TheLong-T ermStructure oftheBlues? Douglas EckandJÄurgenSchmidhuber IstitutoDalleMolledi Studisull'IntelligenzaArti¯ciale(IDSIA) Galleria2,CH6928Manno,Switzerland doug@idsia.ch ,juergen@idsia.ch ,http://www.idsia.ch/ Abstract.Ingeneralmusiccomposedbyrecurrentneuralnetworks

. Current desktop environments provide weak support f or carrying out complex user-oriented tasks. Although individual applications are becoming increasingly sophisticate d and feature- rich, users must map their high-level goals to the low-level opera- tional vocabulary of applications, and deal with a myriad of rou- tine tasks (such as keeping up with email,

Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation Kyunghyun Cho Bart van Merri ¨enboer Caglar Gulcehre Universit ´e de Montr ´eal firstname.lastname@umontreal.caDzmitry Bahdanau Jacobs University, Germany d.bahdanau@jacobs-university.de Fethi Bougares Holger Schwenk Universit ´e du Maine,

The Case for Learned Index Structures Tim Kraska∗ MIT Cambridge, MA kraska@mit.eduAlex Beutel Google, Inc. Mountain View, CA alexbeutel@google.comEd H. Chi Google, Inc. Mountain View, CA edchi@google.com Jeffrey Dean Google, Inc. Mountain View, CA jeff@google.comNeoklis Polyzotis Google, Inc. Mountain View, CA

AI agents are an exciting new research direction, and agent development is driven by benchmarks. Our analysis of current agent benchmarks and evaluation practices reveals several shortcomings that hinder their usefulness in real-world applications. First, there is a narrow focus on accuracy without attention to other metrics. As a result, SOTA

ADVANCES AND CHALLENGES IN FOUNDATION AGENTS FROM BRAIN -INSPIRED INTELLIGENCE TO EVOLUTIONARY , COLLABORATIVE ,AND SAFE SYSTEMS Bang Liu2,3,20∗†, Xinfeng Li4∗, Jiayi Zhang1,10∗, Jinlin Wang1∗, Tanjin He5∗, Sirui Hong1∗, Hongzhang Liu6∗,Shaokun Zhang7∗,Kaitao Song8∗,Kunlun Zhu9∗,Yuheng

Despite the rapid growth of machine learning research, corresponding code imple- mentations are often unavailable, making it slow and labor-intensive for researchers to reproduce results and build upon prior work. In the meantime, recent Large Lan- guage Models (LLMs) excel at understanding scientific documents and generating high-quality code.

of the research on anomaly detection. We have grouped existing techniques into different categoriesbased on the underlying approach adopted by each technique. For each category we have identified key as-sumptions, which are used by the techniques to differentiate between normal and anomalous behavior. Whenapplying a given technique to a

of the research on anomaly detection. We have grouped existing techniques into different categoriesbased on the underlying approach adopted by each technique. For each category we have identified key as-sumptions, which are used by the techniques to differentiate between normal and anomalous behavior. Whenapplying a given technique to a

In this paper we investigate the use of deep neural networks (DNNs) for a small footprint text-dependent speaker verification task. At de- velopment stage, a DNN is trained to classify speakers at the frame- level. During speaker enrollment, the trained DNN is used to extract speaker specific features from the last hidden layer. The average of these

Despite significant recent advances in the field of face recognition [10, 14, 15, 17], implementing face verification and recognition efficiently at scale presents serious chal- lenges to current approaches. In this paper we present a system, called FaceNet, that directly learns a mapping from face images to a compact Euclidean space where distances directly

Designing appropriate features for acoustic event recogni- tion tasks is an active field of research. Expressive features should both improve the performance of the tasks and also be interpret-able. Currently, heuristically designed features based on the domain knowledge requires tremendous effort in hand-crafting, while features

Learning filter widths of spectral decompositions with wavelets Haidar Khan Department of Computer Science Rensselaer Polytechnic Institute Troy, NY 12180 khanh2@rpi.eduBülent Yener Department of Computer Science Rensselaer Polytechnic Institute Troy, NY 12180 yener@rpi.edu Abstract Time series classification using deep neural

The availability of open-source software is playing a remarkable role in the popularization of speech recognition and deep learning. Kaldi, for instance, is nowadays an established framework used to develop state-of-the-art speech recognizers. PyTorch is used to build neural networks with the Python language and has recently spawn tremen- dous

Information Retrieval P. BAXENDALE, Editor A Relational Model of Data for Large Shared Data Banks E. F. CODD IBM Research Laboratory, San Jose, California Future users of large data banks must be protected from having to know how the data is organized in the machine (the internal representation). A prompting service

Inaninformation system akeyroleisplayed bytheunderlying data schema. This article starts out from theviewthattheentire modelling process ofaninformation system' sdata schema canbeseen asa schema transformation process. Atransformation process thatstarts outwith aninitial draft conceptual schema andends with aninternal database schema

The Lustre Storage Architecture Peter J. Braam (with others) Cluster File Systems, Inc. http://www.clusterfs.com 09/29/2005, HEAD

Database Management Systems (DBMSs) are a ubiquitous and critical component of modern computing, and the result of decades of researchand development in both academia and industry. Historically, DBMSswere among the earliest multi-user server systems to be developed, andthus pioneered many systems design techniques for scalability and relia-

17Analysis of Fork/Join and Related Queueing Systems ALEXANDER THOMASIAN , Thomasian and Associates Fork/join (F/J) requests arise in contexts such as parallel computing, query processing in parallel databases, and parallel disk access in RAID. F/J requests spawn Ktasks that are sent to Kparallel servers, and the completion of all

!"$#% &' (*),+-. $/0 ,21354)6"$)879;:=<?> @?:A2<CB$D 9;:E<FBHG*@?:=D I'JLKNM?OPJRQSKNBLT U V2@XW<FBLTYDD[Z]\=^`_a_cb`_edNf;_gh`i`jlknm`o$Z]\pknqrm`\rqcsaZ]kng_mt`uwvxlqpuNyPm`knzqruwxwknj|{s}L~ fP~ns`f;vm`v`v`s`e^`vm?shqpk lknvm?se;vm`o`P\rxp~ x|dif?~ \rv qph`vuljwgqrmjj?_edNf;_gh`i`jwqru Z]\rknqrm`\pq...

9/7/2015 Iceberg-Cubes http://www2.cs.uregina.ca/~dbd/cs831/notes/dcubes/iceberg.html 1/7SELECT FROM CUBE BY HAVING A,B,C, Count(*),SUM(X) TableName A,B,C COUNT(*)>=minsup PartStoreLocationCustomer P1 Vancouver Vance P1 Calgary Bob P1 Toronto Richard P2 Toronto AllisonCombination Count {P1, ANY, ANY} 3 {P2, ANY, ANY} 4

Multidimensional analysis requires the computation of many aggregate functions over a large volume of collected data. To provide the various viewpoints for the analysts, these data are organized as a multi -dimensional data model called data cubes. Each cel l in a data cube represents a unique set of values for the different dimensions and contains the

High-Dimensional OLAP: AMinimal Cubing Approac h¤ Xiaolei Li JiaweiHan HectorGonzalez UniversityofIllinoisatUrbana-Champaign, Urbana, IL61801,USA Abstract Datacubehasbeenplayinganessentialrole infastOLAP(onlineanalytical processing) inmanymulti-dimensional datawarehouses.

of pattern clustering methods from a statistical pattern recognition perspective,with a goal of providing useful advice and references to fundamental conceptsaccessible to the broad community of clustering practitioners. We present ataxonomy of clustering techniques, and identify cross-cutting themes and recentadvances. We also describe some

This report summarizes the First International Workshop on Novel Data Stream Pattern Mining held at the 16th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, on July 25 2010 in Washington, DC.

Sampling vs Sketching: An Information Theoretic Comparison Paul Tune and Darryl Veitch Department of Electrical and Electronic Engineering University of Melbourne, Australia Email:{lsptune@ee., dveitch@ }unimelb.edu.au Abstract —The main approaches to high speed measurement in routers are traffic sampling, and sketching. However, it is not known which paradigm is inherently better at extracting inf

Big Data is identified by its three Vs, namely velocity, volume, and variety. The area of data stream processing has long dealt with the former two Vsvelocity and volume. Over a decade of intensive research, the community has provided many important research discoveries in the area. The third Vof Big Data has been the result of social media and the large unstructured datait generates. Streaming tec

Every day, huge volumes of sensory, transactional, and web data are continuously generated as streams, which need to be analyzedonline as they arrive. Streaming data can be considered as one of the main sources of what is called big data. While predictive modeling for data streams and big data have received a lot of at-tention over the last decade, many research approaches are typically designed

Truth discovery is a long-standing problem for assessing the va- lidity of information from various data sources that may provide different and con flicting information. With the increasing promi- nence of data streams arising in a wide range of applications such as weather forecast and stock price prediction, effective techniques for truth discovery in data streams are demanded. However, exist- in

of Data Exploration Techniques Stratos Idreos Harvard University stratos@seas.harvard.eduOlga Papaemmanouil Brandeis University olga@cs.brandeis.eduSurajit Chaudhuri Microsoft Research surajitc@microsoft.com ABSTRACT Data exploration is about efficiently extracting knowledge from data even if we do not know exactly what we are look- ing for. In this tutorial, we survey recent developments in the area

Datasheets for Datasets Timnit Gebru∗1, Jamie Morgenstern2, Briana Vecchione3, Jennifer Wortman Vaughan4, Hanna Wallach4, Hal Daumé III4,5, and Kate Crawford4,6 1Google 2Georgia Institute of Technology 3Cornell University 4Microsoft Research 5University of Maryland 6AI Now Institute April 16, 2019 Abstract The machine learning community currently has no standardized process for documenting dataset

1 Survey of Graph Database Models RENZO ANGLES and CLAUDIO GUTIERREZ Universidad de Chile Graph database models can be defined as those in which data structures for the schema and instances are modeled as graphs or generalizations of them, and data manipulation is expressed by graph-orientedoperations and type constructors. These models took off in the eighties and early nineties alongside object-o

Query languages for graph databases started to be inves- tigated some 25 years ago. With much current data, suchas linked data on the Web and social network data, be-ing graph-structured, there has been a recent resurgencein interest in graph query languages. We provide a brief survey of many of the graph query languages that have been proposed, focusing on the core functionality provided in th

Scalability in Model-Driven Engineering (MDE) is often a bot- tleneck for industrial applications. Industrial scale models need to be per- sisted in a way that allows for their seamless and efficient manipulation, often by multiple stakeholders simultaneously. This paper compares the conventional and commonly used persistence mechanisms in MDE with novel approaches such as the use of graph-based NoS

Graph databases attempt to make the modelling and processing of highly interconnected data, easier and more efficient by representing a system as a graph-like structure of nodes and edges. The fundamental premise of Graph Databases , unlike relationaL is explicit and distinct definition of relationships and direct, non- index based access of relakd nodes from a given node. Growth of graph database

Gremlin is a graph traversal machine and language designed, de- veloped, and distributed by the Apache TinkerPop project. Grem- lin, as a graph traversal machine, is composed of three interacting components: a graph, a traversal, and a set of traversers. The tra- versers move about the graph according to the instructions spec- ified in the traversal, where the result of the computation is the ultim

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! " "#$% &('*),+-#. %/10,2!#/" !34 ! 506#87" % :9<;=6>@?>AB#.C%) DFEG'*)('H#. I*JLKNM,O%=PRQTS*UHVXWZYU.[>@J%']\'*^*_._"`.acbedFfHgchHbi`8a1jZgXbikl`.anmo p)q#r %'*st2p'H#. (% ! %uv3$#I*J% ! %'1wi x& sDyu z"7I*2!#)()(|{ZIH#&n }% o7" %C%2}'H3~)H6>5J%'-3#IJ% ! %'5I*% %I*'H7%&z"#2p2p

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Up dated Octob er /1/0/, /1/9/9/9C/5/./1/./3 Decision T ree Disco v ery Ron Koha vi Ross Quinlanronnyk/@CS/.Stanford/.EDU quinlan/@cse/.unsw/.edu/.auData Mining Sc ho ol of Computer Science and EngineeringBlue Martini Soft w are Sam uels Building/, G/0/8/2/6/0/0 Campus Dr/. Suite /1/7/5 Univ ersit y of New South Wales San Mateo/, CA /9/4/4/0/2 Sydney /2/0/5/0 AustraliaKeywords/ classi/ cation/

. The technology for building knowledge-based systems by inductive inference from examples has been demonstrated successfully in several practical applications. This paper summarizes an approach to synthesizing decision trees that has been used in a variety of systems, and it describes one such system, ID3, in detail. Results from recent studies show ways in which the methodology can be modified t

4 Classification: Basic Concepts,Decision Trees, andModel Evaluation Classification, which is the task of assigning objects to one of several predefined categories, is a pervasive problem that encompasses many diverse applications. Examples include detecting spam email messages based upon the message header and content, categorizing cells as malignant or benign based upon the results of MRI scans, an

HAN 01-fm-i-vi-9780123814791 2011/6/1 3:29 Page i #1Data MiningThird Edition HAN 01-fm-i-vi-9780123814791 2011/6/1 3:29 Page ii #2 The Morgan Kaufmann Series in Data Management Systems (Selected Titles)Joe Celko's Data, Measurements, and Standards in SQLJoe CelkoInformation Modeling and Relational Databases,2nd EditionTerry Halpin, Tony Morgan Joe Celko's Thinking in Sets Joe CelkoBusiness Metadata B

DataCleaning: ProblemsandCurrentApproaches Erhard Rahm /A3HongHai Do UniversityofLeipzig,Germany http://dbs.uni-leipzig.de Abstract We classify data quality problems that are addressed by data cleaning and provide an overview of the main solution approaches. Data cleaning is especially required when integrating heterogeneous data sources and should be addressed together with schema related data tra

—Often, in the real world, entities have two or more representations in databases. Duplicate records do not share a common key and/or they contain errors that make duplicate matching a difficult task. Errors are introduced as the result of transcription errors,incomplete information, lack of standard formats, or any combination of these factors. In this paper, we present a thorough analysis of the

s and copies of some full reports may be found at: http://cs.anu.edu.au/techreports/ Recent reports in this series: TR-CS-06-01 Stephen M Blackburn, Robin Garner, Chris Hoffmann, Asjad M Khan, Kathryn S McKinley, Rotem Bentzur, Amer Diwan, Daniel Feinberg, Daniel Frampton, Samuel Z Guyer, Martin Hirzel, Antony Hosking, Maria Jump, Han Lee, J Eliot B Moss, Aashish Phansalkar, Darko Stefanović, Th

description of each methodology. Categories and Subject Descriptors: A.1 [Introductory and Survey]; H.2.m [Database Management]: Miscellaneous General Terms: Management, MeasurementAdditional Key Words and Phrases: Data quality, data quality measurement, data quality assessment, data quality improvement, methodology, information system, quality dimension ACM Reference Format: Batini, C., Cappi,

See discussions, stats, and author profiles for this publication at: http://www.researchgate.net/publication/222668699 Automating the approximate record-matching process ARTICLE in INFORMATION SCIENCES · AUGUST 2002 Impact Factor: 3.89 · DOI: 10.1016/S0020-0255(00)00013-X · Source: DBLP CITATIONS 45 DOWNLOADS 152 VIEWS 137 3 AUTHORS, INCLUDING: Vassilios S. Verykios Hellenic Open University 106 P

We introduce HoloClean, a framework for holistic data repairing driven by probabilistic inference. HoloClean unifies qualitative data repairing, which relies on integrity constraints or external data sources, with quantitative data repairing methods, which leverage statistical properties of the input data. Given an inconsistent dataset as input, HoloClean automatically generates a probabilistic pro

DATAVALIDATION FOR MACHINE LEARNING Eric Breck1Neoklis Polyzotis1Sudip Roy1Steven Euijong Whang2Martin Zinkevich1 ABSTRACT Machine learning is a powerful tool for gleaning knowledge from massive amounts of data. While a great deal of machine learning research has focused on improving the accuracy and efficiency of training and inference algorithms, there is less attention in the equally important e

Overton: A Data System for Monitoring and Improving Machine-Learned Products Christopher R´ e AppleFeng Niu ApplePallavi Gudipati AppleCharles Srisuwananukorn Apple September 13, 2019 Abstract We describe a system called Overton, whose main design goal is to support engineers in building, monitoring, and improving production machine learning systems. Key challenges engineers face are monitoring fin

Variational autoencoders (V AEs), as well as other generative models, have been shown to be efficient and ac- curate for capturing the latent structure of vast amounts of complex high-dimensional data. However, existing V AEs can still not directly handle data that are heterogenous (mixed continuous and discrete) or incomplete (with missing data at random), which is indeed common in real-world appl

Modelsand Issues in DataStreamSystems Brian Babcock ShivnathBabu MayurDatar Rajeev Motwani Jennifer Widom DepartmentofComputerScience StanfordUniversity Stanford, CA 94305babcock,shivnath,datar,rajeev,widom @cs.stanford.edu Abstract In this overview paper we motivate the need for and research issues arising from a new model of data processing. In this model, data does not take the form of pers

CQL,aContinuous Query Language,issupportedbytheSTREAM prototypeDataStream Managemen tSystematStanford. CQLisanexpressiveSQL-based declarativ elanguage for registering continuousqueriesagainststreamsandupdatablerelations. Webeginbypresenting

Thespace complexity ofapproximating thefrequency moments NogaAlon*YossiMatias t MarioSzegedy $ Abstract Thefrequency moments ofasequence containing m~ele- mentsoftypei,for1~z~n,arethenumbers F&= ~~=1m?.Weconsider thespacecomplexity ofrandomized algorithms thatapproximate thenumbers Fk,whentheel- ements of the sequence are given one by one and cannot be stored. Surprisingly, it turns out the numbers F0, F1,

ContinuousQueriesoverDataStreams ShivnathBabuandJenniferWidom StanfordUniversityshivnath,widom@cs.stanford.edu http://www-db.stanford.edu/stream Abstract Inmanyrecentapplications,datamaytaketheformof continuousdatastreams,ratherthanfinitestoreddata sets. Severalaspectsofdatamanagementneedtobere- considerediftheprescenceofdatastreams offeringanew

ModelsandIssuesinDataStreamSystems BrianBabcock ShivnathBabuMayurDatarRajeevMotwaniJenniferWidom DepartmentofComputer Science StanfordUniversity Stanford,CA94305babcock ,shivnath,d atar,rajeev ,widom @cs.stanford .edu Abstract Inthisoverviewpaperwemotivatetheneedforandresearchissuesarisingfromanewmodelof dataprocessing. Inthismodel,datadoesnottaketheformofpersistent relations, butratherarrives

Applications that require real-time processing of high-volume data steams are pushing the limits of traditional data processing infrastructures. These stream-based applications include market feed processing and electronic trading on Wall Street, network and infrastructure monitoring, fraud detection, and command and

Given a data stream of length nover an alphabet [m] where n > m, we consider the problem of finding a duplicate in a single pass. We give a randomized algorithm for this problem that uses O((log m)3) space. This answers a question of Muthukrishnan [Mut05] and Tarui [Tar07], who asked if this problem could be solved using sub-linear space and one pass over the input. Our algorithm solves the more ge

This paper describes a unifi cation of two different SQL extensions for streams and its associated semantics. We use the data models from Oracle and StreamBase as our examples. Oracle uses a time-based execution model while StreamBase uses a tuple-based execution model. Time-based execution provides a way to model simultaneity while tuple-based execution provides a way to react to primitive events.

IT 10 013 Examensarbete 15 hp Maj 2010 Evaluation of the Stream Query Language CQL Robert Kajic Institutionen för informationsteknologi Department of Information Technology

IT 10 013 Examensarbete 15 hp Maj 2010 Evaluation of the Stream Query Language CQL Robert Kajic Institutionen för informationsteknologi Department of Information Technology

Outlier detection is a well established area of statistics but mostoftheexistingoutlierdetectiontechniquesaredesigned for applications where the entire dataset is available for ran- dom access. A typical outlier detection technique constructs a standard data distribution or model and identifies the de- viated data points from the model as outliers. Evidently these techniques are not suitable for an

Chimera: A Declarative Language for Streaming Network Traffic Analysis Kevin Borders National Security Agency krborde@tycho.nsa.govJonathan Springer Reservoir Labs springer@reservoir.comMatthew Burnside National Security Agency msburns@tycho.nsa.gov Abstract Intrusion detection systems play a vital role in network security. Central to these systems is the language used to express policies. Ideally

Discretized Streams: A Fault-Tolerant Model for Scalable Stream Processing Matei Zaharia Tathagata Das Haoyuan Li Timothy Hunter Scott Shenker Ion Stoica Electrical Engineering and Computer Sciences University of California at Berkeley Technical Report No. UCB/EECS-2012-259

15Processing Flows of Information: From Data Stream to Complex Event Processing GIANPAOLO CUGOLA and ALESSANDRO MARGARA , Politecnico di Milano A large number of distributed applications requires continuous and timely processing of information as it flows from the periphery to the center of the system. Examples include intrusion detection systems which analyze network traffic in real time to identify

of the usually employed experimental methodologies. Anumber of references are provided that describe applications of data stream clustering in different domains,such as network intrusion detection, sensor networks, and stock market analysis. Information regardingsoftware packages and data repositories are also available for helping researchers and practitioners. Finally,some important issues and c

of the usually employed experimental methodologies. Anumber of references are provided that describe applications of data stream clustering in different domains,such as network intrusion detection, sensor networks, and stock market analysis. Information regardingsoftware packages and data repositories are also available for helping researchers and practitioners. Finally,some important issues and c

Most of the applications in some special domains such as Telecommunication systems, Share market, Fraud detection and network security which required online processing of incoming data. They produce very high incoming load which needs to process by multiple nodes. The current system is on single node bottleneck and with the static configuration and hence it is not able to scale with the input load

MillWheel is a framework for building low-latency data-processing applications that is widely used at Google. Users specify a directed computation graph and application code for individual nodes, and the system manages persistent state and the continuous flow of records, all within the envelope of the framework's fault-tolerance guarantees. This paper describes MillWheel's programming model as well

This paper describes the use of Storm at Twitter. Storm is a real - time fault- tolerant and distributed stream data processin g system . Storm is currently being used to run various critical computations in Twitter at scale , and in real - time. This paper describes the architecture of Storm and its methods for distributed scale -out and fault-tolerance. This paper also describes how queries (jobs

In this tutorial we present the results of recent research about the cloud enablement of data streaming systems. We illus- trate, based on both industrial as well as academic prototypes, new emerging uses cases and research trends. Specifically, we focus on novel approaches for (1) scalability and (2) fault tolerance in large scale distributed streaming systems. In general, new fault tolerance mech

Forrester Research, Inc., 60 Acorn Park Drive, Cambridge, MA 02140 USA Tel: +1 617.613.6000 | Fax: +1 617.613.5000 | www.forrester.comThe Forrester Wave™: Big Data Streaming Analytics Platforms, Q3 2014 by Mike Gualtieri and Rowan Curran, July 17, 2014For: Application Development & Delivery ProfessionalsKey TaKeaways exploiting Perishable Insights to a Huge, Untapped Opportunity For Firms Forrester

The Dataflow Model: A Practical Approach to Balancing Correctness, Latency, and Cost in Massive-Scale, Unbounded, Out-of-Order Data Processing Tyler Akidau, Robert Bradshaw, Craig Chambers, Slava Chernyak, Rafael J. Fern ´andez-Moctezuma, Reuven Lax, Sam McVeety, Daniel Mills, Frances Perry, Eric Schmidt, Sam Whittle Google {takidau, robertsh, chambers, chernyak, rfernand, relax, sgmc, millsd, fip

The Dataflow Model: A Practical Approach to Balancing Correctness, Latency, and Cost in Massive-Scale, Unbounded, Out-of-Order Data Processing Tyler Akidau, Robert Bradshaw, Craig Chambers, Slava Chernyak, Rafael J. Fern ´andez-Moctezuma, Reuven Lax, Sam McVeety, Daniel Mills, Frances Perry, Eric Schmidt, Sam Whittle Google {takidau, robertsh, chambers, chernyak, rfernand, relax, sgmc, millsd, fip

In recent years, our customers have expressed frustration in the traditional approach of using a combination of disparate products to handle their streaming, transactional and ana- lytical needs. The common practice of stitching heteroge- neous environments in custom ways has caused enormous production woes by increasing development complexity and total cost of ownership. With SpanneData, an open

Twitter Heron: Towards Extensible Streaming Engines Karthik Ramasamyt, Ashvin Agrawalm, Avrilia Floratoum, Maosung Fut, Bill Grahamt, Andrew Jorgensent Mark Lit, Neng Lut, Sriram Raom, Cong Wangt tTwitter, Inc.{@karthikz, @Louis Fumaosong, @billgraham, @andrew, @objclone, @luneng90, @c0ngwang } mMicrosoft{asagr, avflor, sriramra }@microsoft.com Abstract — Twitter's data centers process billions of

Ribbon filter: practically smaller than Bloom and Xor1 Peter C. Dillinger Facebook, Inc. Seattle, Washington, USA peterd@fb.comStefan Walzer University of Cologne Cologne, Germany walzer@cs.uni-koeln.de ABSTRACT Filter data structures over-approximate a set of hashable keys, i.e. set membership queries may incorrectly come out positive. A

NATIONAL CENTER FOR EDUCATION STATISTICS Working Paper Series The Working Paper Series was initiated to promote the sharing of the valuable work experience and knowledge reflected in these preliminary reports. These reports are viewed as works in progress, and have not undergone a rigorous review for consistency with NCES

CHAPTER 25 Missing-data imputation Missing data arise in almost all serious statistical analys es. In this chapter we discuss a variety of methods to handle missing data, includi ng some relatively simple approaches that can often yield reasonable results. We use a s a running example the Social Indicators Survey, a telephone survey of New York Cit y

This paper presents the reshape package for R, which provides a common framework for many types of data reshaping and aggregation. It uses a paradigm of ‘melting’ and ‘casting’, where the data are ‘melted’ into a form which distinguishes measured and iden- tifying variables, and then ‘cast’ into a new shape, whether it be a data frame, list, or

Missing data is a common problem in real-world settings and for this reason has attracted significant attention in the statistical literature. We propose a flexible framework based on formal optimization to impute missing data with mixed continuous and categorical variables. This framework can readily incorporate various predictive models including K-

"!# $%& '()*+, -/. $0, 132547698"2;:=<?>47@A25BC8ED3FGHJILKNMPO9O9ORQ 25@TSVU;WX25YZSV[C472>25\C47[D34J]_^a`CbTMPO9O9Odce8"6fBC@Tg25B_IhD347YjikMmlfb?^3n9n9nbCopC4 7@Aq$MPOb?^3n9nMmrsut+vxwxyz|{7w}5~55e7; x[illegible]7PJJ5Ze,Z[illegible]7=[illegible]C ,“e7e5 'P [illegible] xe[illegible] '[illegible]5...

BaggingPredictors By LeoBreiman* Technical ReportNo.421 September 1994 *Partiallysupported byNSFgrantDMS-9212419 Department ofStatistics University ofCalifornia Berkeley, California 94720 ! "$#&%('*),+-/.103254 6879.;:<6>=@?>AB6"256 C&DE68C&F1DG!:CIH*.;4JDKCL6KM(=@?>NO2@PICQ?R=,48:SC&2 256!T>.;4

Fifty years have passed since the publication of the first regression tree algorithm. New tech- niques have added capabilities that far surpass those of the early methods. Modern classification trees can partition the data with linear splits on subsets of variables and fit nearest neighbor, kernel density, and other models in the partitions. Regression

An increasing number of domains are providing us with detailed trace data on human decisions in settings where we can evaluate the quality of these decisions via an algorithm. Motivated by this de- velopment, an emerging line of work has begun to consider whether we can characterize and predict the kinds of decisions where people are likely to make

Computer says “no”: automation, algorithms and artificial intelligence in Government decision-making* Dominique Hogan-Doran SC† Automating systems can assist individuals making administrative decisions or help identify relevant criteria, evidence or particular issues for consideration. Automation has the potential to

—In this paper we propose a synergistic melting of neural networks and decision trees (DT) we call neural decision trees (NDT). NDT is an architecture a la decision tree where each splitting node is an independent multilayer perceptron allowing oblique decision functions or arbitrary nonlinear decision func- tion if more than one layer is used. This way,

..... 3

XAI Metrics p. 1 Metrics for Explainable AI: Challenges and Prospects Robert R. Hoffman Institute for Human and Machine Cognition [rhoffman@ihmc.us] Shane T. Mueller Michigan Technological University [shanem@mtu.edu] Gary Klein Macrocognition, LLC [gary@macrocognition.com] Jordan Litman Institute for Human and Machine Cognition

Combining Ensemble Learning and Human Expertise to Engineer Better Scorecards Gerald Fahner FICO 181 Metro Drive, Suite 700 San Jose, CA 95110 USA +1 (408) 535-1500 geraldfahner@fico.com ABSTRACT Credit scoring applications are often based on the flexible, interpretable, and constrainable (in terms of functional form) model family of

Intelligent artificial agents need to be able to explain and justify their actions. They must therefore understand the rationales for their own actions. This paper describes a technique for acquiring this understanding, implemented in a multimedia explanation system. The system de- termines the motivation for a decision by recall- ing the situation in

TURING AWARD LECTURE Reflections on Trusting Trust To what extent should one trust a statement that a program is free of Trojan horses? Perhaps it is more important to trust the people who wrote the software. KEN THOMPSON INTRODUCTION I thank the ACM for this award. I can't help but feel that I am receiving this honor for timing and

The ethical issues related to the possible future creation of machines with general intellectual capabilities far outstripping those of humans are quite distinct from any ethical problems arising in current automation and information systems. Such superintelligence would not be just another technological development; it would be the most important

As the artificial intelligence (AI) systems in military simulations and computer games become more complex, their actions become increasingly difficult for users to understand. Expert systems for medical diagnosis have addressed this challenge though the addition of explanation generation systems that explain a system's internal processes. This paper

/BT /CA/CT/DA/CX/CT/DB /D3/CU /BX/DC/D4/D0/CP/D2/CP/D8/CX/D3/D2 /C5/CT/D8/CW/D3 /CS/D7/CU/D3/D6 /BU/CP /DD /CT/D7/CX/CP/D2 /C6/CT/D8 /DB /D3/D6/CZ/D7/BV/CP/D6/D1/CT/D2 /C4/CP/CR/CP /DA /CT/BW/CT/D4/D8/BA /BV/D3/D1/D4/D9/D8/CT/D6 /CB/CR/CX/CT/D2/CR/CT/CD/D2/CX/DA /CT/D6/D7/CX/D8 /DD

. We present an overview of different theories of explanation from the phi- losophy and cognitive science communities. Based on these theories, as well as models of explanation from the knowledge-based systems area, we present a framework for explanation in case-based reasoning (CBR) based on explanation goals. We propose ways that the goals

As artificial intelligence (AI) systems and behavior models in military simulations become increasingly complex,ithasbeendifficultforuserstounderstandthe activitiesofcomputer-controlledentities. Prototypeex- planation systems have been added to simulators, but designers have not heeded the lessons learned from work in explaining

. The ability to give explanations for its reasoning and behaviour is a core capability of an intelligent system. There are a number of different goals a user can have towards such explanations. This paper presents how the knowledge intensive case-based reasoning framework CREEK can support some of these different goals in an ambient intelligence setting.

. We propose a hybrid model based on self-regulatory feedback. It is a connectionist ne twork, which can be composed in a modular fashion. It can be com posed of simple constructs that can be combined together to interact logically without requiring a- priori declaration of 'interaction-ty pe' connections. It is primarily a classifier but resolves binding a

After building a classifier with modern tools of machine lear ning we typically have a black box at handthatisabletopredictwellforunseendata. Thus,weget ananswertothequestion whatisthe mostlikelylabelofagivenunseendatapoint. However,most methodswillprovidenoanswer why the model predicted a

There is a common problem in artificial intel- ligence (AI) and information security. In AI, an expert system needs to be able to justify and explain a decision tothe user. In information security, experts need to be able to explain to the public why a system is secure. In both cases, an important goal of explanation is to acquire or maintainthe users' trust. In this

H I S T O R I E S A N D F U T U R E S 2 1541-1672/11/$26.00 © 2011 IEEE IEEE INTELLIGENT SYSTEMS Published by the IEEE Computer SocietyEditor: robert r. hoffman, Jeffrey M. Bradshaw, and Kenneth M. Ford, Institute for Human and Machine Cognition, rhoffman@ihmc.usH U M A N - C E N T E R E D C O M P U T I N G Improving Users' Mental

Recently, ontology stream reasoning has been introduced as a multidisciplinary approach, merging synergies from Artifi- cial Intelligence, Database and World-Wide-Web to reason on semantics-augmented data streams, thus a way to answering questions on real time events. However existing approaches do not consider stream change diagnosis i.e.,

—This paper presents a framework for unsupervised learning of a hierarchical reconfigurable image template — the AND-OR Template (AOT) for visual objects. The AOT includes: (1) hierarchical composition as "AND" nodes, (2) deformation and articulation of parts as geometric "OR" nodes, and (3) multiple ways of composition as structural "OR"

Making machine learning models interpretable Alfredo Vellido1,J o s ´eD .M a r t ´ ın-Guerrero2 and Paulo J.G. Lisboa3∗ 1- Dept. de Llenguatges i Sistemes Inform` atics - Univ. Polit` ecnica de Catalunya C. Jordi Girona, 1-3, 08034 Barcelona - Spain 2- Dept. d'Enginyeria Electr` onica - Universitat de Val` encia Av. de la Universitat, sn, 46100

Standard generalized additive models (GAMs) usually model the dependent variable as a sum of univariate models. Al- though previous studies have shown that standard GAMs can be interpreted by users, their accuracy is significantly less than more complex models that permit interactions. In this paper, we suggest adding selected terms of inter- acting

Building Interpretable Classifiers with Rules using Bayesian Analysis Benjamin Letham MITCynthia Rudin MIT Tyler H. McCormick University of WashingtonDavid Madigan Columbia University Technical Report no. 609 Department of Statistics University of Washington December, 2012 Abstract We aim to produce predictive

RESEARCH ARTICLE EXPLAINING DATA-DRIVEN DOCUMENT CLASSIFICATIONS1 David Martens Department of Engineering Management, Faculty of Applied Economics, University of Antwerp, Prinsstraat 13, 2018 Antwerp, BELGIUM {david.martens@ua.ac.be} Foster Provost Department of Information, Operations and Management Sci

We propose a learning algorithm for fair clas- sification that achieves both group fairness (the proportion of members in a protected group receiving positive classification is iden- tical to the proportion in the population as a whole), and individual fairness (similar in- dividuals should be treated similarly). We formulate fairness as an optimization prob- lem

Intelligent systems encompass a wide range of software technologies including heuristic and normative expert systems, case -based reasoning systems, and neural networks. This field has been augmented in recent years by Web -based applications, such as recommender systems and the semantic Web. The us es of explanation facilities have their roots

—Research is emerging on how end users can correct mistakes their intelligent agents make, but before users can correctly “debug” an intelligent agent, they need some degree of understanding of how it works. In this paper we consider ways intelligent agents should explain themselves to end users, especially focusing on how the soundness and completeness of the explanations impacts the fidelity of

Interpretable classifiers using rules and Bayesian analysis: Building a better stroke prediction model Benjamin Letham Cynthia Rudin MIT MIT Tyler H. McCormick David Madigan University of Washington Columbia University Technical Report no. 609 Department of Statistics University of Washington August 2013

OF THE DISSERTATION OF Todd Kulesza for the degree of Doctor of Philosophy in Computer Science presented on December 1, 2014. Title: Personalizing Machine Learning Systems with Explanatory Debugging Abstract approved: Margaret M. Burnett How can end users efficiently influence the predictions that machine learning systems make on their behalf? Traditional systems rely on users to provide examples of

RESEARCH ARTICLES◥ COGNITIVE SCIENCE Human-level concept learning through probabilisticprogram induction Brenden M. Lake,1*Ruslan Salakhutdinov,2Joshua B. Tenenbaum3 People learning new concepts can often generalize successfully from just a single example, yet machine learning algorithms typically require tens or hundreds of examples to perform with similar accuracy. People can also use learned ca

In machine learning often a tradeoff must be made between accuracy and intelligibility. More accurate models such as boosted trees, random forests, and neural nets usually are not intelligible, but more intelligible models such as logistic regression, naive-Bayes, and single decision trees often have significantly worse accuracy. This tradeoff sometimes limits the accuracy of models that can be applie

The Annals of Applied Statistics 2015, V ol. 9, No. 3, 1350–1371 DOI: 10.1214/15-AOAS848 ©Institute of Mathematical Statistics , 2015 INTERPRETABLE CLASSIFIERS USING RULES AND BAYESIAN ANALYSIS: BUILDING A BETTER STROKE PREDICTION MODEL BYBENJAMIN LETHAM∗,1,CYNTHIA RUDIN∗,1,TYLER H. M

How can end users efficiently influence the predictions that machine learning systems make on their behalf? This paper presents Explanatory Debugging, an approach in which the system explains to users how it made each of its predictions, and the user then explains any necessary corrections back to the learning system. We present the principles underlying this approach and a prototype instantiating i

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privacy and rightsacmqueue | november-december 2015 1Every fiscal quarter automated writing algorithms churn out thousands of corporate earnings articles for the AP (Associated Press) based on little more than structured data. Companies such as Automated Insights, which produces the articles for AP, and Narrative Science can now write straight news articles in almost any domain that has clean and

A Step Towards Accountable Algorithms?: Algorithmic Discrimination and the European Union General Data Protection Bryce W. Goodman Oxford Internet Institute University of Oxford bwgoodman@gmail.com Abstract Algorithms, and the data they process, play an increasingly important role in deci- sions with significant consequences for human welfare. This trend has given rise to calls for greater accounta

Deep models are the defacto standard in visual decision models due to their impressive performance on a wide array of visual tasks. However, they are frequently seen as opaque and are unable to explain their decisions. In contrast, hu- mans can justify their decisions with natural language and point to the evidence in the visual world which led to their decisions. We postulate that deep models can

Broad Agency Announcement Explainable Artificial Intelligence (XAI) DARPA-BAA-16-53 August 10, 2016 Defense Advanced Research Projects Agency Information Innovation Office 675 North Randolph Street Arlington, VA 22203-2114

We summarize the potential impact that the European Union’s new General Data Protection Regulation will have on the routine use of machine learning algorithms. Slated to take effect as law across the EU in 2018, it will restrict automated individual decision-making (that is, algorithms that make decisions based on user- level predictors) which “significantly affect” users. The law will also effecti

Explainable Agency for Intelligent Autonomous Systems Pat Langley∗andBen Meadows∗andMohan Sridharan‡ Department of Computer Science∗/ Electrical and Computer Engineering‡ University of Auckland, Private Bag 92019, Auckland 1142, NZ Dongkyu Choi Department of Aerospace Engineering University of Kansas, Lawrence, KS 66045 USA Explainable Agency As intelligent agents become more autonomous, complexi

1 Explainable Artificial Intelligence (XAI) David Gunning DARPA/I2O Distribution Statement "A" (Approved for Public Release, Distribution Unlimited) Watson AlphaGo 2 Explainable AI – What Are We Trying To Do? User AI System Sensemaking Operations •Why did you do that? •Why not something else? •When do you succeed? •When do you fail? •When can I trust you? •How do I correct an error? •We are enter

Generating Visual Explanations Lisa Anne Hendricks1Zeynep Akata2Marcus Rohrbach1,3 Jeff Donahue1Bernt Schiele2Trevor Darrell1 1UC Berkeley EECS, CA, United States 2Max Planck Institute for Informatics, Saarbr¨ ucken, Germany 3ICSI, Berkeley, CA, United States Abstract. Clearly explaining a rationale for a classification decision to an end-user can be as important as the decision itself. Existing en

In recent years there is a growing interest in us- ing deep representations for reinforcement learn- ing. In this paper, we present a methodology and tools to analyze Deep Q-networks (DQNs) in a non-blind matter. Using our tools we re- veal that the features learned by DQNs aggregate the state space in a hierarchical fashion, explain- ing its success.

Learning Modular Neural Network Policies for Multi-Task and Multi-Robot Transfer Coline Devin∗1Abhishek Gupta∗1Trevor Darrell1Pieter Abbeel1Sergey Levine1 Abstract — Reinforcement learning (RL) can automate a wide variety of robotic skills, but learning each new skill requires considerable real-world data collection and

Nothing Else Matters: Model-Agnostic Explanations By Identifying Prediction Invariance Marco Tulio Ribeiro University of Washington Seattle, WA 98105 marcotcr@cs.uw.eduSameer Singh University of California, Irvine Irvine, CA 92697 sameer@uci.eduCarlos Guestrin University of Washington Seattle, WA 98105 guestrin@cs.uw.edu 1 Introduction At the

—Machine learning (ML) models, e.g., state-of-the-art deep neural networks (DNNs), are vulnerable to adversarial examples: malicious inputs modified to yield erroneous model outputs, while appearing unmodified to human observers. Po- tential attacks include having malicious content like malware identified as legitimate

Understanding predictive models, in terms of interpreting and identifying actionable insights, is a challenging task. Often the importance of a feature in a model is only a rough esti- mate condensed into one number. However, our research goes beyond these na ¨ıve estimates through the design and imple- mentation of an interactive visual analytics system,

Understanding predictive models, in terms of interpreting and identifying actionable insights, is a challenging task. Often the importance of a feature in a model is only a rough esti- mate condensed into one number. However, our research goes beyond these na ¨ıve estimates through the design and imple- mentation of an interactive visual analytics system,

Verbalization: Narration of Autonomous Robot Experience Stephanie Rosenthal Software Engineering Institute srosenthal@sei.cmu.eduSai P. Selvaraj Robotics Institute spandise@andrew.cmu.eduManuela Veloso Computer Science Department veloso@cs.cmu.edu Carnegie Mellon University, Pittsburgh USA Abstract Autonomous mobile

Despite widespread adoption, machine learning models re- main mostly black boxes. Understanding the reasons behind predictions is, however, quite important in assessing trust, which is fundamental if one plans to take action based on a prediction, or when choosing whether to deploy a new model. Such understanding also provides insights into the model,

123Machine Learning ISSN 0885-6125Volume 106Number 7 Mach Learn (2017) 106:1039-1082DOI 10.1007/s10994-017-5633-9Optimal classification treesDimitris Bertsimas & Jack Dunn

3/23/2017 Is AI making credit scores better , or more confusing? https://www .americanbanker .com/news/is-ai-making-credit-scores-better -or-more-confusing 1/11Is AI making credit scores better, or more confusing? By Penny Crosman Published February 1 4 20 17, 3:35pm ES T More in Artif icial int elligence , Credit scores, Credit quali ty, Bank t

4/14/2017 The Dark Secret at the Heart of AI - MIT T echnology Review https://www .technologyreview .com/s/604087/the-dark-secret-at-the-heart-of-ai/ 1/28L KEITH RANKIN ast year, a strange self-driving car was released onto the quiet roads of Monmouth County, New Jersey. The experimentalIntelligent Machines The Dark Sec ret at the

7/9/2017 Ideas on interpreting machine learning - O'Reilly Media https://www .oreilly .com/ideas/ideas-on-interpreting-machine-learning 1/80 Ideas on interpreting machine learning Mix-and-match approaches for visualizing data and interpreting machine learning models and results. By Patrick Hall, Wen Phan, and SriSatish Ambati. March 15, 2017

78 1541-1672/17/$33.00 © 2017 IEEE IEEE INTELLIGENT SYSTEMS Published by the IEEE Computer Society HUMAN-CENTERED COMPUTING Editor: Robert R. Hoffman, Institute for Human and Machine Cognition, rhoffman@ihmc.usExplaining Explanation, Part 2: Empirical Foundations Robert R. Hoffman, Institute for Human and

As autonomous intelligent systems become more widespread, society is beginning to ask: “What are the machines up to?”. Various forms of artificial intelligence control our latest cars,load balance components of our power grids, dictate much of the movement in our stock markets and help doctors diagnose and treat our ailments. As they become

iv Chapter 1 Introduction 1

Accountability of AI Under the Law: The Role of Explanation Finale Doshi-Velez*, Mason Kortz*, for the Berkman Klein Center Working Group on Explanation and the Law: Ryan Budish, Berkman Klein Center for Internet and Society at Harvard University Chris Bavitz, Harvard Law School; Berkman Klein Center for Internet and Society at Harvard

3 Introduction 6 Labor and Automation 7 Research by Sector and Task 7 AI and the Nature of Work 9 Inequality and Redistribution 1 3 Bias and Inclusion 1 3 Where Bias Comes From 1 4 The AI Field is Not Diverse 1 6 Recent Developments in Bias Research 1 8 Emerging Strategies ...

. Programming is a form of dialogue with machines. In recent years, we have become increasingly involved in a dialogue that shapes our surroundings, as we come to inhabit a newly inferred world. It is unclear how this dialogue should be structured, especially as the notion of “correctness” for these programs is now unknown or ill-defined. I

Current deep neural networks (DNNs) are “black boxes ” 2?data prediction •Do not expose their decision making process •Do not provide their confidence in their predictions •Not clear whether they can be trusted and/or corrected How do we make Deep Learning moreexplainable and trustworthy ?

Deeply Explainable Artificial Intelligence Profs. Trevor Darrell, Dan Klein, Pieter Abbeel, John Canny, Tom Griffiths, Anca Dragan, UC Berkeley; Prof. Kate Saenko, BU; Prof. Zeynep Akata, U. Amsterdam; Dr. Anthony Hoogs, Kitware Our proposed effort toward deeply explainable AI (DEXAI) involves two key challenges: (1) generating accurate

'=2903469 1 Why a right to explanation of automated decision -making does not exist in the G eneral Data Protection Regulation

principles, are to be applied to all soci - etal contexts. Given that the issue of ethics is always also about specific kinds of action and about responsibility for this action, however, it is inevitably dependent on structural and situ - ational contextualization. The rules that apply to the state or to a government, for example, can hardly be applied to citizens as

Quick reference FICORXpress OptimizationGuide for evaluators 2 Advanced optimization tasks Last update 17 May, 2017 www.fico.com Make every decision countTM Evaluator guide 2 (Advanced) FICORXpress Optimization Guide for evaluators 2 Advanced optimization tasks 17 May, 2017 Introduction Building on the introductory Guide for evaluators

– With the availability of large databases and recent improvements in deep learning methodology, the performance of AI systems is reaching, or even exceeding, the human level on an increasing number of complex tasks. Impressive examples of this development can be found in domains such as image

Explainable Planning Maria Fox, Derek Long, Daniele Magazzeni King’s College London firstname.lastname@kcl.ac.uk Abstract As AI is increasingly being adopted into applica- tion solutions, the challenge of supporting inter- action with humans is becoming more apparent. Partly this is to support integrated working styles, in which humans and intelligent

There has been a recent resurgence in the area of explainable artificial intelligence as researchers and practitioners seek to provide more transparency to their algorithms. Much of this research is focused on explicitly explaining decisions or actions to a human observer, and it should not be controversial to say that, if these techniques are to

We propose a technique for making Convolutional Neural Network (CNN)-based models more transparent by visual- izing the input regions that are ‘important’ for predictions from these models – producing visual explanations . Our approach, called Gradient-weighted Class Activation Mapping (Grad-CAM), uses the class-specific gradient infor-

Machine-learned models are o/f_ten described as “black boxes”. In many real-world applications however, models may have to sacri- /f_ice predictive power in favour of human-interpretability. When this is the case, feature engineering becomes a crucial task, which requires signi/f_icant and time-consuming human effort. Whilst some features are

Deep neural networks have been shown to be very successful at learning feature hierarchies in supervised learning tasks. Generative models, on the other hand, have benefited less from hierar- chical models with multiple layers of latent vari- ables. In this paper, we prove that hierarchical latent variable models do not take advantage of the

The purported “black box” nature of neural networks is a barrier to adoption in applica- tions where interpretability is essential. Here we present DeepLIFT (Deep Learning Impor- tant FeaTures), a method for decomposing the output prediction of a neural network on a spe- cific input by backpropagating the contributions of all neurons in the network to

Despite their great success, there is still no comprehensive theoretical understanding of learning with Deep Neural Networks (DNNs) or their inner organization. Previous work [Tishby and Za- slavsky (2015)] proposed to analyze DNNs in the Information Plane ; i.e., the plane of the Mutual Information values that each layer preserves on the input and

Plan Explicability and Predictability for Robot Task Planning Yu Zhang1, Sarath Sreedharan1, Anagha Kulkarni1, Tathagata Chakraborti1, Hankz Hankui Zhuo2 and Subbarao Kambhampati1 Abstract — Intelligent robots and machines are becoming per- vasive in human populated environments. A desirable capability of these agents is to

Neural networks are among the most accurate su- pervised learning methods in use today. However, their opacity makes them difficult to trust in critical applications, especially if conditions in training may differ from those in test. Recent work on explana- tions for black-box models has produced tools (e.g. LIME) to show the implicit rules behind

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Towards A Rigorous Science of Interpretable Machine Learning Finale Doshi-Velez∗and Been Kim∗ From autonomous cars and adaptive email-filters to predictive policing systems, machine learn- ing (ML) systems are increasingly ubiquitous; they outperform humans on specific tasks [Mnih et al., 2013, Silver et al., 2016, Hamill, 2017] and often

Towards A Rigorous Science of Interpretable Machine Learning Finale Doshi-Velez∗and Been Kim∗ From autonomous cars and adaptive email-filters to predictive policing systems, machine learn- ing (ML) systems are increasingly ubiquitous; they outperform humans on specific tasks [Mnih et al., 2013, Silver et al., 2016, Hamill, 2017] and often

Understanding why a model makes a certain prediction can be as crucial as the prediction’s accuracy in many applications. However, the highest accuracy for large modern datasets is often achieved by complex models that even experts struggle to interpret, such as ensemble or deep learning models, creating a tension between accuracy

Understanding why a model makes a certain prediction can be as crucial as the prediction’s accuracy in many applications. However, the highest accuracy for large modern datasets is often achieved by complex models that even experts struggle to interpret, such as ensemble or deep learning models, creating a tension between accuracy

Patrick Hall & Navdeep GillAn Applied Perspective on Fairness, Accountability, Transparency, and Explainable AIAn Introduction to Machine Learning Interpretability Compliments ofdataiku

93A Survey of Methods for Explaining Black Box Models RICCARDOGUIDOTTI,ANNAMONREALE, SALVATORERUGGIERI,and FRANCOTURINI, KDDLab, Universityof Pisa,Italy FOSCA GIANNOTTI, KDDLab, ISTI-CNR,Italy DINOPEDRESCHI, KDDLab, Universityof Pisa,Italy In recent years, many accurate decision support systems

Deep k-Nearest Neighbors: Towards Confident, Interpretable and Robust Deep Learning Nicolas Papernot and Patrick McDaniel Department of Computer Science and Engineering, Pennsylvania State University {ngp5056, mcdaniel }@cse.psu.edu Abstract —Deep neural networks (DNNs) enable innovative applications of machine

Artificial intelligence (AI) generally and machine learning (ML) specifically demonstrate impressive practical success in many different application domains, e.g. in autonomous driving, speech recognition, or recommender systems. Deep learning approaches, trained on extremely large data sets or using reinforcement learning methods have even

We argue that robustness of explanations—i.e., that similar inputs should give rise to similar explanations—is a key desideratum for inter- pretability. We introduce metrics to quantify ro- bustness and demonstrate that current methods do not perform well according to these metrics. Finally, we propose ways that robustness can be enforced on existing

36 COMMUNICATIONS OF THE ACM | OCTOBER 2018 | VOL. 61 | NO. 10practice SUPERVISED MACHINE-LEARNING models boast remarkable predictive capabilities. But can you trust your model? Will it work in deployment? What else can it tell you about the world? Models should be not only good, but also interpretable, yet the task of

The nascent field of fair machine learning aims to ensure that decisions guided by algorithms are equitable. Over the last several years, three formal definitions of fairness have gained promi- nence: (1) anti-classification, meaning that protected attributes—like race, gender, and their proxies—are not explicitly used to make decisions; (2) classification

Code of Practice CODE OF PRACTICE 2007CODE OF PRACTICE 2007CODE OF PRACTICE 2007CODE OF PRACTICE 2007CODE OF PRACTICE 2007CODE OF PRACTICE 2007CODE OF PRACTICE 2007CODE OF PRACTICE 2007CODE OF PRACTICE 2007 CODE OF PRACTICE 2007CODE OF PRACTICE 2007CODE OF

68 COMMUNICATIONS OF THE ACM | JANUARY 2020 | VOL. 63 | NO. 1review articles MACHINE LEARNING IS progressing at an astounding rate, powered by complex models such as ensemble models and deep neural networks (DNNs). These models have a wide range of real-world applications, such as movie recommendations of Netflix, neural

Model Cards for Model Reporting Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, Timnit Gebru {mmitchellai,simonewu,andrewzaldivar,parkerbarnes,lucyvasserman,benhutch,espitzer,tgebru}@google.com

Predict Responsibly: Improving Fairness and Accuracy by Learning to Defer David Madras, Toniann Pitassi & Richard Zemel University of Toronto Vector Institute {madras,toni,zemel}@cs.toronto.edu Abstract In many machine learning applications, there are multiple decision-makers involved, both automated and human. The interaction between these

We present a machine learning pipeline for fairness-aware machine learning (FAML) in fi- nance that encompasses metrics for fairness (and accuracy). Whereas accuracy metrics are well un- derstood and the principal ones used frequently, there is no consensus as to which of several avail- able measures for fairness should be used in a generic

44 COMMUNICATIONS OF THE ACM | AUGUST 2021 | VOL. 64 | NO. 8practice IMAGE BY SUPERPUAY/SHUTTERSTOCKA CHILD WEARING sunglasses is labeled as a “failure, loser, nonstarter, unsuccessful person.” This is just one of the many systemic biases exposed by ImageNet Roulette, an art

DECODING TRUST : A Comprehensive Assessment of Trustworthiness in GPT Models Boxin Wang1∗, Weixin Chen1∗, Hengzhi Pei1∗, Chulin Xie1∗, Mintong Kang1∗, Chenhui Zhang1∗, Chejian Xu1, Zidi Xiong1, Ritik Dutta1, Rylan Schaeffer2, Sang T. Truong2, Simran Arora2, Mantas Mazeika1, Dan Hendrycks3,4, Zinan Lin5, Yu

Classification Simply put, the goal of classification is to determine a plausible value for an unknown variable Ygiven an observed variable X. For exam- ple, we might try to predict whether a loan applicant will pay back her loan by looking at various characteristics such as credit history, income, and net worth. Classification also applies in situations where the

Introduction Our success, happiness, and wellbeing are never fully of our own making. Others' decisions can profoundly affect the course of our lives: whether to admit us to a particular school, offer us a job, or grant us a mortgage. Arbitrary, inconsistent, or faulty decision- making thus raises serious concerns because it risks limiting our ability to achieve the goals that we have set for ours

Causality Our starting point is the difference between an observation and an action. What we see in passive observation is how individuals follow their routine behavior, habits, and natural inclination. Passive ob- servation reflects the state of the world projected to a set of features we chose to highlight. Data that we collect from passive observation show a snapshot of our world as it is. There

—Learning visual models of object categories notoriously requires hundreds or thousands of training examples. We show that it is possible to learn much information about a category from just one, or a handful, of images. The key insight is that, rather than learning from scratch, one can take advantage of knowledge coming from previously learned categories, no matter how different these categories

One of the main challenges in feature learning using Deep Convolutional Neural Networks (DCNNs) for large- scale face recognition is the design of appropriate loss func- tions that enhance discriminative power. Centre loss pe- nalises the distance between the deep features and their cor- responding class centres in the Euclidean space to achieve intra-class compactness. SphereFace assumes that the

Generalizing from a Few Examples: A Survey on Few-Shot Learning YAQING WANG, Hong Kong University of Science and Technology and 4Paradigm Inc QUANMING YAO, 4Paradigm Inc JAMES T. KWOK, Hong Kong University of Science and Technology LIONEL M. NI, Hong Kong University of Science and Technology Artificial intelligence succeeds in data-intensive applications, but it lacks the ability of learning from

A. M. Turing (1950) Computing Machinery and Intellig ence. Mind 49: 433-460. COMPUTING MACHINERY AND INTELLIGENCE By A. M. Turing 1. The Imit ation Gam e I propose to consider the question, "Can m achines think? " This should begin with definition s of the m eaning of the ter ms "machine" and " think." The def

! "#$ %#& #' #&"( ) #&*" +-,/.(0$132$46587:9<;-=2>4856?A@CB-567(1!@CDEDGFIHCF.,KJL,/.NMEO/PRQS9<;UT2$46V&2$4XW%Y OUMEV<FI4ZP[ME5R9\],/^_@:0X7-F>PR58FI4';-`X,ba,c.,d1!@&48e/@C4Z2$56ME@CO1f,bg ,Kh<7/2>O-O-@COi;KajFIDGDIkmFIDGFIe-7-@&O-

nIn recent years, there has been a flurry of research into empirical, corpus-based learning approachesto natural language processing (NLP). Most empir-ical NLP work to date has focused on relatively low-level language processing such as part-of- speech tagging, text segmentation, and syntacticparsing. The success of these approaches has stim-ulated research in using empirical learning techniques i

. This paper shows how a dialogue system for information-seeking dialogues can be imple- mented in a type-theory-based syntax editor, originally developed for editing mathematical proofs.The implementation gives a simple logical metatheory to such dialogue systems and also suggests new functions for them, e.g., a local undo operation. The method developed provides a logically based declarative way

Computers understand very little of the meaning of human language. This profoundly limits our ability to give instructions to computers, the ability of computers to explain their actions to us, and the ability of computers to analyse and process text. Vector space models (VSMs) of semantics are beginning to address these limits. This paper surveys the use of VSMs for semantic processing of text. W

"9781405155816_4_011" — 2010/5/8 — 11:59 — page 271 — #111 Evaluation of NLP SystemsPHILIP RESNIK AND JIMMY LIN1I n t r o d u c t i o nAs the engineering branch of computational linguistics, natural language process-ing is concerned with the creation of artifacts that accomplish tasks. The operativequestion in evaluating an NLP algorithm or system is therefore the extent to which it produces the re

of the types of databases. Se ction 4 presents the acoustic characteristics of emotions. Section 5 presents feature extracti on and classification. Section 6 discusses the applications of emotion recognition. Section 7 presents concluding remarks. 2. Basic framework for emotional recognition The input files are speech signals. Fig.1 gi ves the basic framework of emotional speech recognition. The f

Proceedings of the 50th Annual Meeting of the Association for Computational Linguistics , pages 712–720, Jeju, Republic of Korea, 8-14 July 2012. c2012 Association for Computational Linguistics Unsupervised Relation Discovery with Sense Disambiguation Limin Yao Sebastian Riedel Andrew McCallum Department of Computer Science University of Massachusetts, Amherst {lmyao,riedel,mccallum}@cs.umass.ed

The science of AI is concerned with the study of intelligent forms of behaviour in computational terms. But what does it tell us when a good sem- blance of a behaviour can be achieved using cheap tricks that seem to have little to do with what we intuitively imagine intelligence to be? Are these intuitions wrong, and is intelligence really just a bag of tricks? Or are the philosophers right, and i

Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing , pages 1631–1642, Seattle, Washington, USA, 18-21 October 2013. c2013 Association for Computational Linguistics Recursive Deep Models for Semantic Compositionality Over a Sentiment Treebank Richard Socher, Alex Perelygin, Jean Y. Wu, Jason Chuang, Christopher D. Manning, Andrew Y. Ng and Christopher Potts Stan

Over the past few years, neural networks have re-emerged as powerful machine-learning models, yielding state-of-the-art results in fields such as image recognition and speech processing. More recently, neural network models started to be applied also to textual natural language signals, again with very promising results. This tutorial surveys neural network models from the perspective of natural la

Reasoning about Entailment with Neural Attention Tim Rockt ¨aschel University College London t.rocktaschel@cs.ucl.ac.ukEdward Grefenstette & Karl Moritz Hermann Google DeepMind {etg,kmh}@google.com Tom ´a˘s Ko ˘cisk´y University of Oxford tomas@kocisky.euPhil Blunsom Google DeepMind & University of Oxford

Learning Distributed Word Representations for Natural Logic Reasoning Samuel R. Bowman, Christopher Potts, Christopher D. Manning Stanford University 450 Serra Mall Stanford, CA 94305-2150 {sbowman,cgpotts,manning}@stanford.edu Abstract Natural logic offers a powerful relational conception of meaning

Beyond Accuracy: Behavioral Testing of NLP Models with C heckList Marco Tulio Ribeiro Microsoft Research marcotcr@microsoft.comTongshuang Wu Univ. of Washington wtshuang@cs.uw.eduCarlos Guestrin Univ. of Washington guestrin@cs.uw.eduSameer Singh Univ. of California, Irvine sameer@uci.edu Abstract Although

Findings of the Association for Computational Linguistics: EMNLP 2020 , pages 1383–1392 November 16 - 20, 2020. c2020 Association for Computational Linguistics1383Modeling Intra and Inter-modality Incongruity for Multi-Modal Sarcasm Detection Hongliang Pan, Zheng Lin, Peng Fu, Yatao Qi, Weiping Wang Chinese Academy of Sciences

arXiv:2102.02503v1 [cs.CL] 4 Feb 2021Understanding the Capabilities, Limitations, and Societal Impact of Large Language Models Alex Tamkin∗1, Miles Brundage∗2, Jack Clark†3, and Deep Ganguli1,3 1Stanford University2OpenAI3AI Index Introduction On October 14th, 2020, researchers from OpenAI, the Stanfor d Institute for

"! # $&%('*)+-,/.1012 %435+6' 78+9%($:,*);,=< > ?@?*AB)+6' .&C D CFEHG@I +-,/.J +HEKCL<M,@)N.O) CP,QSRNTUTWVYXZ\[UT]^V5_ Xa`T9b\`Ncdcd`[]BQ`Nedf[gRhb\XiTkjNlm`k]gRTknWopRcrqlms tHu RvwTkXiqxgy{z}|~ ¡¢£¤¥¦§¨©ª«¬®¯°±²³´µ¶·¸¹º»¼½¾¿ÀÁÂÃÄÅÆÇÈÉÊËÌÍÎÏÐÑÒÓÔÕÖ×ØÙÚÛÜÝÞßàáâãäåæçèéêëìíîïðñòóôõö÷øùúûüýþÿ...

1 Introduction to Semi-Supervised Learning 1.1 Supervised, Unsupervised, and Semi-Supervised Learning In order to understand the nature of semi-supervised learning, it will be useful first to take a look at supervised and unsupervised learning. 1.1.1 Supervised and Unsupervised LearningTraditionally, there have been two fundamentally different types

of pattern clustering methods from a statistical pattern recognition perspective,with a goal of providing useful advice and references to fundamental conceptsaccessible to the broad community of clustering practitioners. We present ataxonomy of clustering techniques, and identify cross-cutting themes and recentadvances. We also describe some

Using Data Mining Techniques for Fraud Detection Solving Business Problems Using SAS®Enterprise Miner™Software A SAS Institute Best Practices Paper In conjunction with Federal Data Corporation

The setup for measuring the SHG is described in the supporting online material (22). We expect that the SHG strongly depends on the resonance that is excited. Obviously, the incident polariza- tion and the detuning of the laser wavelength from the resonance are of particular interest. One possibility for control ling the detuning is to change the laser

Com bining Lab eled and Unlab eled Data with Co/-T raining / yAvrim BlumSc ho ol of Computer ScienceCarnegie Mellon Univ ersit yPittsburgh/, P A /1/5/2/1/3/-/3/8/9/1avrim/+/@cs/.cmu/.ed u T om Mitc hellSc ho ol of Computer ScienceCarnegie Mellon Univ ersit yPittsburgh/, P A /1/5/2/1/3/-/3/8/9/1mitchell/+/@cs/.cmu/.e duAbstractW e consider

8.LEARNING INTERNAL REPRESENT AnONS319 network withouthiddenunits)willbeunabletoperform thenecessary mappings. Aclassicexample ofthiscaseistheexclusive-or (XOR) problem illustrated inTable1.Hereweseethatthosepatterns which overlapleastaresupposed togenerate identical outputvalues. This problem

ACTA CLIMA TOLOGICA ET CHOROLOGICA Universitatis Szegedie nsis, Tom . 36-37 , 2003 , 45-56. COMPARISON OF ARTIFI CIAL INT ELLIGENCE PREDICTION TECHNIQUES IN NO AND NO2 CONCE NTR ATIONS' FORE CAST I. JUHOS1, R. BÉCZI2 and L. MAKR A2 1Department of Informatics, University of Szeged, P.O.Box 652, 6701 Szeg ed,

Frequent pattern mining has been a focused theme in data mining re- search for over a decade. Abundant literature has been dedicated to this researchand tremendous progress has been made, ranging from efficient and scalable algorithms for frequent itemset mining in transaction databases to numerous research frontiers, such as sequential

Many researchers see the need for reject inference in credit scoring models to come from a sample selection problem whereby a missing variable results in omitted variable bias. Alternatively, practitioners often see the problem as one of missing data where the relationship in the new model is biased because the behaviour of the omitted cases differs from

L ? EXPLORATIONS IN PARALLEL DISTRIBUTED PROCESSING A Handbook of Models, Programs, and Exercises James L. McClelland David E. Rumelhart A Bradford Book The MIT Press Cambridge, Massachusetts London, England ;:,"!/fji~ ~ L ;..f:(j~~':: ~li1j.."".,...-... ? Computational Models of Cognition and Perception Editors Jerome A. Feldman

1 LearningDeepArchitecturesforAI Yoshua Bengio Dept. IRO, Universit´ edeMontr´ eal C.P. 6128,Montreal,Qc, H3C3J7,Canada Yoshua.Bengio@umontreal.ca http://www.iro.umontreal.ca/ ~bengioy To appearin FoundationsandTrends inMachineLearning Abstract Theoreticalresultssuggest thatinorder tolearnthekindo

Semi-Supervised Learning Xiaojin Zhu, University of Wisconsin-Madison Synonyms: Learning from labeled and unlabeled data, transductive learn- ing Definition Semi-supervised learning uses both labeled and unlabeled data to perform an otherwise supervised learning or unsupervised learning task. In the former case, there is a distinction

—A major assumption in many machine learning and data mining algorithms is that the training and future data must be in the same feature space and have the same distribution. However, in many real-world applications, this assumption may not hold. For example, we sometimes have a classi fication task in one domain of interest, but we only have suf

201Why Does Unsupervised Pre-training Help Deep Learning? Dumitru Erhan Aaron Courville Yoshua Bengio Pascal Vincent DIRO, Universit ´e de Montr ´eal 2920 chemin de la Tour Montr ´eal, Qu ´ebec, H3T 1J8, Canada first.last@umontreal.ca Abstract Much recent research has been devoted to learning algorithms for deep

Introduction to Machine Learning Alex Smola and S.V.N. Vishwanathan Yahoo! Labs Santa Clara –and– Departments of Statistics and Computer Science Purdue University –and– College of Engineering and Computer Science Australian National University

78 communications of the acm | october 2012 | vol. 55 | no. 10review articles Machine learning syste Ms automatically learn programs from data. This is often a very attractive alternative to manually constructing them, and in the last decade the use of machine learning has spread rapidly throughout computer science and beyond. Machine learning is

78 communications of the acm | october 2012 | vol. 55 | no. 10review articles Machine learning syste Ms automatically learn programs from data. This is often a very attractive alternative to manually constructing them, and in the last decade the use of machine learning has spread rapidly throughout computer science and beyond. Machine learning is

LONG SHOR T/-TERM MEMOR YNeural Comput a tion /9/(/8/)/:/1/7/3/5/{/1/7/8/0/, /1/9/9/7Sepp Ho c hreiterF akult/ at f / ur InformatikT ec hnisc he Univ ersit/ at M / unc hen/8/0/2/9/0 M / unc hen/, German yho c hreit/@informatik/.tu/-m uenc hen/.deh ttp/:////www/7/.informatik/.tu/-m uenc hen/.de///~ho c hreit J / urgen Sc

Deep Learning in Neural Networks: An Overview Technical Report IDSIA-03-14 / arXiv:1404.7828 v4 [cs.NE] (88 pages, 888 references) J¨urgen Schmidhuber The Swiss AI Lab IDSIA Istituto Dalle Molle di Studi sull'Intelligenza Artificiale University of Lugano & SUPSI Galleria 2, 6928 Manno-Lugano Switzerland 8 October 2014 Abstract In recent years,

. Convolutional neural networks are at the core of state-of- the-art approaches to a variety of computer vision tasks. Visualizationsof neural networks typically take the form of static diagrams, or interac- tive toy-sized networks, which fail to illustrate the networks' scale and complexity, and furthermore do not enable meaningful experimentation.Motivated by

Deep Convolutional Networks on Graph-Structured Data Mikael Henaff Courant Institute of Mathematical Sciences New York University mbh305@nyu.eduJoan Bruna University of California, Berkeley joan.bruna@berkeley.edu Yann LeCun Courant Institute of Mathematical Sciences New York University yann@cs.nyu.edu Abstract Deep Learning's recent

Efficient and Robust Automated Machine Learning Matthias Feurer Aaron Klein Katharina Eggensperger Jost Tobias Springenberg Manuel Blum Frank Hutter Department of Computer Science University of Freiburg, Germany {feurerm,kleinaa,eggenspk,springj,mblum,fh }@cs.uni-freiburg.de Abstract The success of machine learning in a broad

Hidden Technical Debt in Machine Learning Systems D. Sculley, Gary Holt, Daniel Golovin, Eugene Davydov, Todd Phillips {dsculley,gholt,dgg,edavydov,toddphillips }@google.com Google, Inc. Dietmar Ebner, Vinay Chaudhary, Michael Young, Jean-Franc ¸ ois Crespo, Dan Dennison {ebner,vchaudhary,mwyoung,jfcrespo,dennison }@google.com Google,

Hidden Technical Debt in Machine Learning Systems D. Sculley, Gary Holt, Daniel Golovin, Eugene Davydov, Todd Phillips {dsculley,gholt,dgg,edavydov,toddphillips }@google.com Google, Inc. Dietmar Ebner, Vinay Chaudhary, Michael Young, Jean-Franc ¸ ois Crespo, Dan Dennison {ebner,vchaudhary,mwyoung,jfcrespo,dennison }@google.com Google,

. One of the most important assets to be protected is information, as every aspect of the life of a society deeply depends on the available information. Nowadays, information is stored, processed, and communicated by computers. It turns out that computers represent the most critical tool in modern society. A number of protection mechanisms are available so

Google's Neural Machine Translation System: Bridging the Gap between Human and Machine Translation Yonghui Wu, Mike Schuster, Zhifeng Chen, Quoc V. Le, Mohammad Norouzi yonghui,schuster,zhifengc,qvl,mnorouzi@google.com Wolfgang Macherey, Maxim Krikun, Yuan Cao, Qin Gao, Klaus Macherey, Jeff Klingner, Apurva Shah, Melvin Johnson,

June 25th 2016SPECIAL REPORT ARTIFICIAL INTELLIGENCE The return of the machinery question 20160625_AI.indd 1 13/06/2016 19:55 The Economist June 25th 201 6 1ARTIFICIAL INTELLIGENCESPECIAL REPOR T Alist of sources is at Economist.com/specialreportsCONTENTS 2 Technology From not working to neur al networking

Technical Report Number 884 Computer LaboratoryUCAM-CL-TR-884 ISSN 1476-2986 Machine learning and computer algebra Zongyan Huang April 2016 15 JJ Thomson Avenue Cambridge CB3 0FD United Kingdom phone +44 1223 763500 http://www.cl.cam.ac.uk/ c2016 Zongyan Huang This technical report is based on a dissertation submitted

Building a machine learning model is an iterative process. A data scientist will build many tens to hundreds of models before arriving at one that meets some acceptance criteria (e.g. AUC cutoff, accu- racy threshold). However, the current style of model building is ad-hoc and there is no practical way for a data scientist to manage models that are built over

Attention Is All You Need Ashish Vaswani∗ Google Brain avaswani@google.comNoam Shazeer∗ Google Brain noam@google.comNiki Parmar∗ Google Research nikip@google.comJakob Uszkoreit∗ Google Research usz@google.com Llion Jones∗ Google Research llion@google.comAidan N. Gomez∗† University of Toronto

s ;rComputing methodologies → Learning paradigms ;Learning settings ;Machine learning approaches ; Cognitive robotics; Control methods; Distributed artificial intelligence; Computer vision; Additional Key Words and Phrases: Imitation learning, learning from demonstrations, intelligent agents, learning from experience, self-improvement, feature

Since its appearance, Generative Adversarial Net- works (GANs) [2] have received a lot of interest in the AI community. In image generation several projects showed how GANs are able to generate photorealis- tic images but the results so far didn't look adequate for the quality standard of visual media production industry.

ATM: A distributed, collaborative, scalable system for automated machine learning Thomas Swearingen∗, Will Drevo†, Bennett Cyphers†, Alfredo Cuesta-Infante‡, Arun Ross∗and Kalyan Veeramachaneni† ∗Department of Computer Science and Engineering Michigan State University, East Lansing, Michigan 48824 Email:

Model-based Pricing for Machine Learning in a Data Marketplace Lingjiao Chen†, Paraschos Koutris†and Arun Kumar‡ †Department of Computer Science, University of Wisconsin, Madison ‡Department of Computer Science and Engineering, University of California, San Diego Abstract Data analytics using machine learning (ML) has become

Data Platform for Machine Learning Pulkit Agrawal, Rajat Arya, Aanchal Bindal, Sandeep Bhatia, Anupriya Gagneja, Joseph Godlewski, Yucheng Low, Timothy Muss, Mudit Manu Paliwal, Sethu Raman, Vishrut Shah, Bochao Shen, Laura Sugden, Kaiyu Zhao, Ming-Chuan Wu∗ Apple Inc. Cupertino, CA ABSTRACT In this paper, we present a purpose-

TOWARDS FEDERATED LEARNING AT SCALE : SYSTEM DESIGN Keith Bonawitz1Hubert Eichner1Wolfgang Grieskamp1Dzmitry Huba1Alex Ingerman1Vladimir Ivanov1 Chlo ´e Kiddon1Jakub Kone ˇcn´y1Stefano Mazzocchi1H. Brendan McMahan1Timon Van Overveldt1 David Petrou1Daniel Ramage1Jason Roselander1 ABSTRACT Federated

/ C h a p t e r 1 0 Se q u e n ce Mo de l i ng: Recurren t and Recursiv e N e t s Recurrentneuralnetworks ,orRNNs(,),areafamily Rumelhart etal.1986a ofneuralnetworksforprocessingsequentialdata.Muchasaconvolutionalnetwork isaneuralnetworkthatisspecializedforprocessingagridofvalues X suchas

Hidden Technical Debts for Fair Machine Learning in Financial Services Chong Huang* FICO AI Research San Jose, CA 95110 chonghuang@fico.comArash Nourian⇤ FICO AI Research San Rafael, CA 94903 arashnourian@fico.comKevin Griest FICO AI Research San Rafael, CA 94903 kevingriest@fico.com Abstract The

ML models often exhibit unexpectedly poor behavior when they are deployed in real-world domains. We identify underspecification as a key reason for these failures. An ML pipeline is underspecified when it can return many predictors with equivalently strong held-out performance in the training domain. Underspecification is common in

ML models often exhibit unexpectedly poor behavior when they are deployed in real-world domains. We identify underspecification as a key reason for these failures. An ML pipeline is underspecified when it can return many predictors with equivalently strong held-out performance in the training domain. Underspecification is common in

What is being transferred in transfer learning? Behnam Neyshabur∗* Google neyshabur@goog/l.Vare.comHanie Sedghi∗ Google Brain hsedghi@goog/l.Vare.comChiyuan Zhang∗ Google Brain chiyuan@goog/l.Vare.com August /three.prop/one.prop, /two.prop/zero.prop/two.prop/zero.prop Abstract One desired capability for machines is the ability

Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting Haoyi Zhou,1Shanghang Zhang,2Jieqi Peng,1Shuai Zhang,1Jianxin Li,1 Hui Xiong,3Wancai Zhang4 1Beihang University2UC Berkeley3Rutgers University4SEDD Company {zhouhy, pengjq, zhangs, lijx }@act.buaa.edu.cn,

PyGlove: Symbolic Programming for Automated Machine Learning Daiyi Peng, Xuanyi Dong, Esteban Real, Mingxing Tan, Yifeng Lu Hanxiao Liu, Gabriel Bender, Adam Kraft, Chen Liang, Quoc V . Le Google Research, Brain Team {daiyip, ereal, tanmingxing, yifenglu, hanxiaol, gbender, adamkraft, crazydonkey, qvl}@google.com

arXiv:2506.04592v1 [cs.CL] 5 Jun 2025Safe: Enhancing Mathematical Reasoning in Large Language Models via Retrospective Step-aware Formal V erification Chengwu Liu1∗, Ye Yuan1, Yichun Yin2, Yan Xu2, Xin Xu3∗, Zaoyu Chen4∗,Yasheng Wang2,Lifeng Shang2,Qun Liu2,Ming Zhang1, 1School of Computer Science, National Key

In this paper, we describe an architecture for conversational systems that enables human-like performance along several important dimensions. First, interpretation is incremental, multi-level, and involves both general and task- and domain- specific knowledge. Second, generation is also incremental, proceeds in parallel with interpretation, and accounts for phenomena such as turn-taking, grounding

A Computational Architecture for Conversation Eric Horvitz1 and Tim Paek2* 1 Decision Theory and Adaptive Systems, Microsoft Research, WA, U.S.A. 2 Department of Psychology, Stanford University, CA, U.S.A. Abstract. We describe representation, inference strategies, and control procedures em- ployed in an automated

The Mathematics of Statistical Machine Translation: Parameter Estimation Peter E Brown* IBM T.J. Watson Research Center Vincent J. Della Pietra* IBM T.J. Watson Research Center Stephen A. Della Pietra* IBM T.J. Watson Research Center Robert L. Mercer* IBM T.J. Watson Research Center We describe a series o,f five statistical models o,f the translation process and give algorithms for estimating the

of the state of the art. We describe the context of thecurrent research and then move to a formal problem description and an overview of the main subproblems:translation modeling, parameter estimation, and decoding. Along the way, we present a taxonomy of somedifferent approaches within these areas. We conclude with an overview of evaluation and a discussion of future directions. Categories and Sub

[Reproduced by permission of the Rockefeller Foundation Archives.] The attached memorandum on translation from one language to another, and on the possibility of contributing to this process by the use of modern computing devices of very high speed, capacity, and logical flexibility, has been written with one hope only - that it might possibly serve in some small way as a stimulus to someone else

Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics , pages 1491–1500, Baltimore, Maryland, USA, June 23-25 2014. c2014 Association for Computational Linguistics A Recursive Recurrent Neural Network for Statistical Machine Translation Shujie Liu1, Nan Yang2, Mu Li1and Ming Zhou1 1Microsoft Research Asia, Beijing, China 2University of Science and Technology of C

arXiv:1409.3215v3 [cs.CL] 14 Dec 2014Sequenceto SequenceLearning withNeuralNetworks IlyaSutskever Google ilyasu@google.comOriolVinyals Google vinyals@google.comQuocV.Le Google qvl@google.com Abstract Deep Neural Networks (DNNs) are powerful models that have ac hieved excel- lentperformanceondifficultlearningtasks. Although DNNs work well whenever large labeled training sets are available, they cannot

Conversational modeling is an important task in natural language understanding and machine in- telligence. Although previous approaches ex- ist, they are often restricted to specific domains (e.g.,bookinganairlineticket)andrequirehand- crafted rules. In this paper, we present a sim- pleapproachforthistaskwhichusestherecently proposed sequence to sequence framework. Our model conversesby predicting

Google’s Neural Machine Translation System: Bridging the Gap between Human and Machine Translation Yonghui Wu, Mike Schuster, Zhifeng Chen, Quoc V. Le, Mohammad Norouzi yonghui,schuster,zhifengc,qvl,mnorouzi@google.com Wolfgang Macherey, Maxim Krikun, Yuan Cao, Qin Gao, Klaus Macherey, Jeff Klingner, Apurva Shah, Melvin Johnson, Xiaobing Liu, Łukasz Kaiser, Stephan Gouws, Yoshikiyo Kato, Taku Kudo

Google’s Neural Machine Translation System: Bridging the Gap between Human and Machine Translation Yonghui Wu, Mike Schuster, Zhifeng Chen, Quoc V. Le, Mohammad Norouzi yonghui,schuster,zhifengc,qvl,mnorouzi@google.com Wolfgang Macherey, Maxim Krikun, Yuan Cao, Qin Gao, Klaus Macherey, Jeff Klingner, Apurva Shah, Melvin Johnson, Xiaobing Liu, Łukasz Kaiser, Stephan Gouws, Yoshikiyo Kato, Taku Kudo

Published as a conference paper at ICLR 2016 MULTI -TASK SEQUENCE TO SEQUENCE LEARNING Minh-Thang Luong∗, Quoc V . Le, Ilya Sutskever, Oriol Vinyals, Lukasz Kaiser Google Brain lmthang@stanford.edu, {qvl,ilyasu,vinyals,lukaszkaiser }@google.com ABSTRACT Sequence to sequence learning has recently emerged as a new paradigm in supervised learning. To date, most of its applications focused only

Toward Multilingual Neural Machine Translation with Universal Encoder and Decoder Thanh-Le Ha, Jan Niehues, Alexander Waibel Institute for Anthropomatics and Robotics KIT - Karlsruhe Institute of Technology, Germany firstname.lastname@kit.edu Abstract In this paper, we present our first attempts in building a multilingual Neural Machine Translation framework under a unified approach. We are then able

Neural Machine Translation and Sequence-to-sequence Models: A Tutorial Graham Neubig Language Technologies Institute, Carnegie Mellon University 1 Introduction This tutorial introduces a new and powerful set of techniques variously called “neural machine translation” or “neural sequence-to-sequence models”. These techniques have been used in a number of tasks regarding the handling of human language

Neural Machine Translation and Sequence-to-sequence Models: A Tutorial Graham Neubig Language Technologies Institute, Carnegie Mellon University 1 Introduction This tutorial introduces a new and powerful set of techniques variously called “neural machine translation” or “neural sequence-to-sequence models”. These techniques have been used in a number of tasks regarding the handling of human language

Sequence to Sequence Learning with Neural Networks Ilya Sutskever Google ilyasu@google.comOriol Vinyals Google vinyals@google.comQuoc V . Le Google qvl@google.com Abstract Deep Neural Networks (DNNs) are powerful models that have ac hieved excel- lent performance on difficult learning tasks. Although DNNs work well whenever large labeled training sets are available, they cannot be used to map sequ

The fields of machine learning and mathematical programming are increasingly intertwined. Op- timization problems lie at the heart of most machine learnin g approaches. The Special Topic on Machine Learning and Large Scale Optimization examines thi s interplay. Machine learning re- searchershaveembracedtheadvancesinmathematicalprogr amming allowing ever more of models to be pursued. The special topic in

Constraints manuscript No. (will be inserted by the editor) The Future of Optimization Technology Maria Garcia de la Banda ·Peter J. Stuckey · Pascal Van Hentenryck ·Mark Wallace the date of receipt and acceptance should be inserted later Abstract Technology for combinatorial optimization is rapidly changing, and as the size and scope of problems that

The move from hand-designed features to learned features in machine learning has been wildly successful. In spite of this, optimization algorithms are still designed by hand. In this paper we show how the design of an optimization algorithm can be cast as a learning problem, allowing the algorithm to learn to exploit structure in the problems of

. JuMP is an open-source modeling language that allows users to express a wide range of optimization problems (linear, mixed-integer, quadratic, conic-quadratic, semidefinite, and nonlinear) in a high-level, algebraic syntax. JuMP takes advantage of advanced features of the Julia programming language to offer unique functionality while achieving

Mixed Integer Programming (MIP) is one of the most widely used modeling techniques for combinatorial opti- mization problems. In many applications, a similar MIP model is solved on a regular basis, maintaining remark- able similarities in model structures and solution ap- pearances but differing in formulation coefficients. This offers the opportunity for

arXiv:1906.01629v2 [cs.LG] 7 Jun 2019Exact Combinatorial Optimization with Graph Convolutional Neural Networks Maxime Gasse Mila, Polytechnique Montréal maxime.gasse@polymtl.caDidier Chételat Polytechnique Montréal didier.chetelat@polymtl.ca Nicola Ferroni Polytechnique Montréal ferronsnicola@gmail.comLaurent Charlin∗ Mila, HEC

Over the past decade, we have seen tremendous interest in the application of data mining and statistical algorithms, first in research and science and, more recently, across vari- ous industries. This has translated into the development of a myriad of solutions by the data mining community that today impact scientific and business applications alike. How-

To enable complex data-intensive applications such as personalized recommendations, targeted advertising, and intelligent services, the data management community has focused heavily on the design of systems to train complex models on large datasets. Unfortunately, the design of these systems largely ignores a critical component of the overall

56 COMMUNICATIONS OF THE ACM | JULY 2018 | VOL. 61 | NO. 7contributed articles SHEEP AND WOLF ILLUSTRATIONS BY ARN0 MACHINE LEARNING HAS advanced radically over the past 10 years, and machine learning algorithms now achieve human-level performance or better on a number of tasks, including face recognition,31

This paper presents the ‘‘CDAC AI life cycle,’’ a comprehensive life cycle for the design, development, and deployment of artificial intelligence (AI) systems and solutions. It addresses the void of a practical and inclu- sive approach that spans beyond the technical constructs to also focus on the challenges of risk analysis of AI adoption, transferability of

We present a fingerprint matching algorithm that initially identifies the candidate common unique (minutiae)points in both the base and the input images using ratios of relative distances as the comparing function. A tree like structure is then drawn connecting the common minutiae points from bottom up in both the base and the input images. Matching

Identification and Verification of person on the basis of biometric features has become known in our society. When the biometric systems ar e connected to a person’s body remarkably decreases the possibility of fraud. A biometric system basically is a pattern recognition system and to recognize the pattern or the image is not an easy task. From the various

We study, for the first time, automated inference on crim- inality based solely on still face images, which is free of any biases of subjective judgments of human observers. Via supervised machine learning, we build four classifiers (logistic regression, KNN, SVM, CNN) using facial im- ages of 1856 real persons controlled for race, gender, age and facial

Journal of Arti/ cial In telligence Researc h /4 /(/1/9/9/6/) /2/3/7/-/2/8/5 Submitted /9///9/5/; published /5///9/6Reinforcemen t Learning/: A Surv eyLeslie P ac k Kaelbling lpk/@cs/.br o wn/.eduMic hael L/. Littman mlittman/@cs/.br o wn/.eduComputer Scienc e Dep artment/, Box /1/9/1/0/, Br own UniversityPr ovidenc e/, RI

Published as a conference paper at ICLR 2017 SEMI-SUPERVISED KNOWLEDGE TRANSFER FOR DEEPLEARNING FROM PRIVATE TRAINING DATA Nicolas Papernot∗ Pennsylvania State University ngp5056@cse.psu.eduMart ´ın Abadi Google Brain abadi@google.com´Ulfar Erlingsson Google

5/23/2018 How will the GDPR impact machine learning? - O'Reilly Media https://www .oreilly .com/ideas/how-will-the-gdpr -impact-machine-learning 1/12How will the GDPR impact machine learning? Answers to the three most commonly asked questions about maintaining GDPR- compliant machine learning programs. By Andrew Burt. May 16, 2018 Light

5/23/2018 Is there a 'right to explanation' for machine learning in the GDPR? https://iapp.or g/news/a/is-there-a-right-to-explanation-for -machine-learning-in-the-gdpr/ 1/6 Jun 1, 2017Is there a 'right to explanation' for machine learning in the GDPR? Save This () (http://bit.ly/2s47Qxm) Much has been made about the coming effects of the GDPR — from

The technical debt metaphor is widely used to encapsulate numerous software quality problems. The metaphor is at- tractive to practitioners as it communicates to both techni- cal and nontechnical audiences that if quality problems are not addressed, things may get worse. However, it is unclear whether there are practices that move this metaphor

As the capabilities of code large language models (LLMs) continue to expand, their applica- tions across diverse code intelligence domains are rapidly increasing. However, most existing datasets only evaluate limited application domains. To address this gap, we have developed a comprehensive code evaluation dataset FullStack Bench1,2focusing on full-

debug-gym : A Text-Based Environment for Interactive Debugging Xingdi Yuan♠∗Morgane M Moss♠♣Charbel El Feghali♢ Chinmay Singh♡Darya Moldavskaya♡Drew MacPhee♠Lucas Caccia♠ Matheus Pereira♠Minseon Kim♠Alessandro Sordoni♠Marc-Alexandre Côté♠∗ ♠Microsoft Research Montréal♡Microsoft Research NYC ♢McGill

From Code Foundation Models to Agents and Applications: A Comprehensive Survey and Practical Guide to Code Intelligence BUAA-SKLCCSE, Alibaba, ByteDance, M-A-P , BJTU, OPPO, HKUST (GZ), BUPT, TeleAI, Shanghai AI Lab, Manchester, StepFun, UoS, SCU, CASIA, NJU, Kuaishou, HIT, Huawei Cloud, Tencent, Monash/CSIRO, NTU,

TURING AWARD LECTURE Reflections on Trusting Trust To what extent should one trust a statement that a program is free of Trojan horses? Perhaps it is more important to trust the people who wrote the software. KEN THOMPSON INTRODUCTION I thank the ACM for this award. I can't help but feel that I am receiving this honor for timing and

Hacking Blind Andrea Bittau, Adam Belay, Ali Mashtizadeh, David Mazi `eres, Dan Boneh Stanford University Abstract —We show that it is possible to write remote stack buffer overflow exploits without possessing a copy of the target binary or source code, against services that restart after a crash. This makes it possible to hack proprietary closed-

We describe matrix computations available in the cluster programming framework, Apache Spark. Out of the box, Spark provides abstractions and implementations for dis- tributed matrices and optimization routines using these ma- trices. When translating single-node algorithms to run on a distributed cluster, we observe that

The convergence behavior of many distributed machine learn- ing (ML) algorithms can be sensitive to the number of ma- chines being used or to changes in the computing environ- ment. As a result, scaling to a large number of machines can be challenging. In this paper, we describe a new scalable coordinate descent (SCD) algorithm for generalized linear

Synner allows users to generate realistic-looking data. With Synner users can visually and declaratively specify prop- erties of the dataset they wish to generate. Such properties include the domain, and statistical distribution of each field, and relationships between fields. User can also sketch cus- tom distributions and relationships. Synner provides instant

TinyTroupe: An LLM-powered Multiagent Persona Simulation Toolkit Paulo Salem, Robert Sim, Christopher Olsen, Prerit Saxena, Rafael Barcelos, Yi Ding∗ Microsoft Corporation ABSTRACT Recent advances in Large Language Models (LLM) have led to a new class of autonomous agents, renewing and expanding interest in the area. LLM-

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